1 calculus

1.1. Introduction

Calculus is the mathematical study of change and it is concerned with comparing quantities which vary in a non-linear way. “Calculus” is a Latin word which has the meaning "small stones used for counting". That is, understanding something by looking at small pieces. It has two major branches, differential calculus concerning rates of change and slopes of curves, and integral calculus concerning accumulation of quantities and the areas under curves. **Differential Calculus** cuts something into small pieces to find how it changes whereas **Integral Calculus** joins (integrates) the small pieces together to find how much there is. Also, these two branches are related to each other by the fundamental theorem of calculus. Differential and integral calculus are the study of the definition, properties, and applications. The process of finding the derivative is called differentiation and that of an integral is called integration. Calculus has widespread uses in science, economics, and engineering and can solve many problems that algebra alone cannot.

1.2. Engineering Applications of Calculus

Calculus is used in every branch of the physical sciences, actuarial science, computer science, statistics, engineering, economics, business, medicine, demography, and in other fields wherever a problem can be mathematically modeled and an optimal solution is desired. It allows one to go from (non-constant) rates of change to the total change or vice versa.

Physics makes particular use of calculus; all concepts in classical mechanics and electromagnetism are interrelated through calculus. The mass of an object of known density, the moment of inertia of objects, as well as the total energy of an object within a conservative field can be found by the use of calculus.

An example of the use of calculus in mechanics is Newton's second law of motion: The **rate of change** of momentum of a body is equal to the resultant force acting on the body and is in the same direction. It is commonly expressed as Force = Mass × acceleration, it involves differential calculus because acceleration is the time derivative of velocity or second time derivative of trajectory or spatial position. Starting from knowing how an object is accelerating, we use calculus to derive its path.

Also, calculus can be used in conjunction with other mathematical disciplines. For example, it can be used with linear algebra to find the "best fit" linear approximation for a set of points in a domain. It can be used in probability theory to determine the probability of a continuous random variable from an assumed density function. In analytic geometry, the study of graphs of functions, calculus is used to find high points and low points (maxima and minima), slope, concavity and inflection points.

1.3. Curvature

Let A be a fixed point on given curve from where actual distances are measured. Let P and Q are two points on the curve, close to each other such that $AP = S$ and $AQ = S + \Delta S$ Then Arc $PQ = \Delta S$. Let ψ and $\psi + \Delta\psi$ be the angles made by the tangents at P and Q respectively with the x-axis. The angle between the tangent at P and $Q = \Delta\psi$. For a length ΔS of the curve, change in the direction of the tangent $= \Delta\psi$

The rate at which this angle has changed in the interval $PQ = \dfrac{\Delta\psi}{\Delta S}$. The rate at which the curve has bent in the interval PQ is $\dfrac{\Delta\psi}{\Delta S}$. The rate of bending of a curve in any interval is called the curvature of the curve in that interval. Thus, curvature at $P = \underset{\Delta s \to 0}{Lt} \dfrac{\Delta\psi}{\Delta S} = \dfrac{d\psi}{dS}$

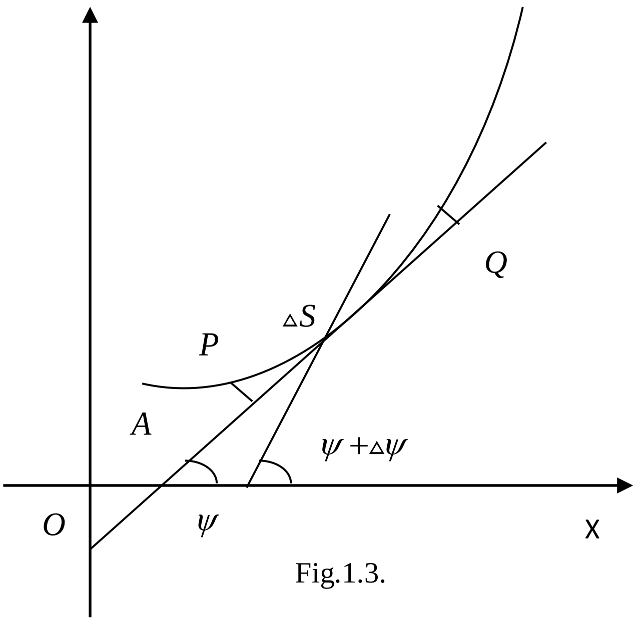

Fig.1.3.

1.4. Radius of curvature

The reciprocal of curvature is defined as radius of curvature.

Note: 1. For a straight line radius of curvature will be ∞.

2. For a circle the radius of curvature will be constant.

1.4.1. Cartesian formula for the radius of curvature

Let P be any point (x,y) on a given curve and ψ be the angle made by the tangent at P with the x-axis. We know that $\tan\psi = \dfrac{dy}{dx}$ …(1)

If ρ is the radius of curvature at P, $\rho = \frac{ds}{d\psi}$ …(2)

Now $\cos\psi = \cos\angle QPR$

$$= \underset{\Delta x \to 0}{\mathrm{lt}} \frac{\Delta x}{PQ}$$

$$= \underset{\Delta x \to 0}{\mathrm{lt}} \frac{\Delta x}{\Delta s}\frac{\Delta s}{PQ}$$ [Multiply and divide by Δs]

$$= \left(\frac{dx}{ds}\right)1$$ [$\frac{\Delta S}{PQ} = 1$ as Q tends to P]

$$\therefore \cos\psi = \frac{dx}{ds}$$ ...(3)

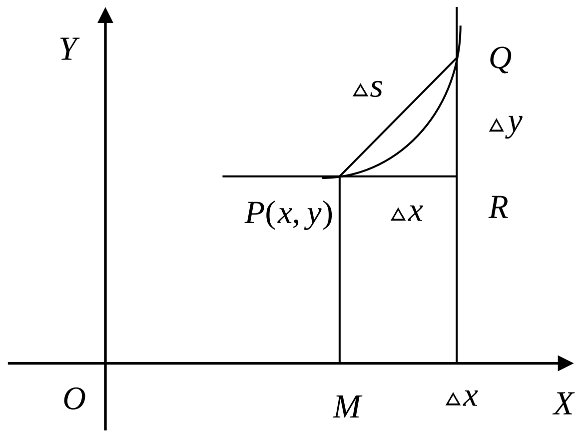

Fig. 1.2.

From (1), $\tan\psi = \frac{dy}{dx}$

Differentiating with respect to x,

$$\sec^2\psi \frac{d\psi}{dx} = \frac{d^2y}{dx^2}$$

$$\sec^2\psi \frac{d\psi}{ds}\frac{ds}{dx} = \frac{d^2y}{dx^2}$$

$$\sec^2\psi \frac{1}{\rho}\frac{1}{\cos\psi} = \frac{d^2y}{dx^2} \qquad \left[\because \frac{ds}{d\psi} = \rho, \frac{dx}{ds} = \cos\psi\right]$$

$$\sec^3\psi \frac{1}{\rho} = \frac{d^2y}{dx^2}$$

$$\rho\frac{d^2y}{dx^2} = \sec^3\psi$$

$$= (\sec^2\psi)^{3/2}$$

$$= (1+\tan^2\psi)^{3/2}$$

$$\rho\frac{d^2y}{dx^2} = \left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}$$

$$\rho = \frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\left(\frac{d^2y}{dx^2}\right)}$$

$$\therefore \rho = \frac{\left[1+y_1^2\right]^{3/2}}{y_2} \quad \text{where} \quad y_1 = \frac{dy}{dx}, \quad y_2 = \frac{d^2y}{dx^2}$$

Note: The modified formula $\rho = \frac{\left[1+\left(\frac{dx}{dy}\right)^2\right]^{3/2}}{\left(\frac{d^2x}{dy^2}\right)}$ can be applied when $\frac{dy}{dx} = \infty$ or $\frac{d^2y}{dx^2} = 0$

1.4.2. Radius of curvature in polar coordinates

From the above figure $\psi = \theta + \phi$

Therefore $\dfrac{d\psi}{d\theta} = 1 + \dfrac{d\phi}{d\theta}$

$$\frac{d\psi}{ds} \cdot \frac{ds}{d\theta} = 1 + \frac{d\phi}{d\theta}$$

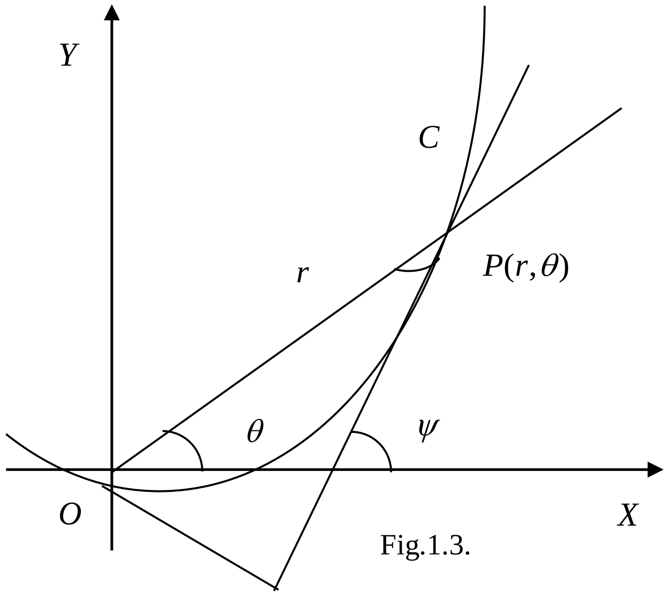

Fig.1.3.

$$\frac{1}{\rho} \cdot \frac{ds}{d\theta} = 1 + \frac{d\phi}{d\theta} \qquad \ldots(1)$$

$$\sin\phi = r\frac{d\theta}{ds},\ \cos\phi = \frac{dr}{ds}$$

We have $\tan\varphi = r\dfrac{d\theta}{dr} = \dfrac{r}{\left(\dfrac{dr}{d\theta}\right)} = \dfrac{r}{r'}$ …(2)

Differentiating (2) with respect to θ θ

$$\sec^2\varphi \frac{d\varphi}{d\theta} = \frac{r'^2 - rr''}{r'^2}$$

$$\frac{d\varphi}{d\theta} = \frac{r'^2 - rr''}{r'^2 \sec^2\varphi}$$

$$= \frac{r'^2 - rr''}{r'^2(1+\tan^2\varphi)} = \frac{r'^2 - rr''}{r'^2(1+\frac{r^2}{r'^2})}$$

$$= \frac{r'^2 - rr''}{r'^2 + r^2} \qquad \ldots(3)$$

Substituting (3) in (1) we get,

$$\frac{1}{\rho}\frac{ds}{d\theta} = 1 + \frac{r'^2 - rr''}{r'^2 + r^2} = \frac{r^2 + 2r'^2 - rr''}{r'^2 + r^2}$$

...(4)

From the length of arc of a curve in polar coordinates,

$$\left(\frac{ds}{d\theta}\right)^2 = r^2 + \left(\frac{dr}{d\theta}\right)^2$$

Therefore $\dfrac{ds}{d\theta} = \sqrt{r^2 + \left(\dfrac{dr}{d\theta}\right)^2} = \sqrt{(r^2 + r'^2)}$ ...(5)

Substituting (5) in (4)

$$\frac{1}{\rho}\sqrt{(r^2 + r'^2)} = \frac{r^2 + 2r'^2 - rr''}{r'^2 + r^2} \qquad \rho = \frac{(r^2 + r'^2)^{3/2}}{r^2 + 2r'^2 - rr''}$$

$$\rho = \frac{\left[r^2 + \left(\frac{dr}{d\theta}\right)^2\right]^{3/2}}{r^2 + 2\left(\frac{dr}{d\theta}\right)^2 - r\left(\frac{d^2r}{d\theta^2}\right)} \qquad \left[\because r' = \frac{dr}{d\theta}, r'' = \frac{d^2r}{d\theta^2}\right]$$

Formula to find radius of curvature:-

1. $\rho = \dfrac{\left[1+\left(\dfrac{dy}{dx}\right)^2\right]^{3/2}}{\left(\dfrac{d^2y}{dx^2}\right)}$ in Cartesian coordinates

2. $\rho = \dfrac{\left[1+\left(\dfrac{dx}{dy}\right)^2\right]^{3/2}}{\left(\dfrac{d^2x}{dy^2}\right)}$ can be applied when $\dfrac{dy}{dx} = \infty$ or $\dfrac{d^2y}{dx^2} = 0$

3. $\rho = \dfrac{\left[r^2+\left(\dfrac{dr}{d\theta}\right)^2\right]^{3/2}}{r^2+2\left(\dfrac{dr}{d\theta}\right)^2 - r\left(\dfrac{d^2r}{d\theta^2}\right)}$ in polar coordinates

$\left[\because r' = \dfrac{dr}{d\theta}, r'' = \dfrac{d^2r}{d\theta^2}\right]$

Example 1: Find the radius of curvature for the curve $\sqrt{x}+\sqrt{y}=1$ **at the point** $\left(\dfrac{1}{4},\dfrac{1}{4}\right)$

. **(May-2008, Nov-2011)**

Solution:

Given that $\sqrt{x}+\sqrt{y}=1$ ...(1)

Differentiating (1) with respect to x, we get

$$\frac{1}{2\sqrt{x}}+\frac{1}{2\sqrt{y}}\frac{dy}{dx}=0\ ;\ \frac{dy}{dx}=-\frac{\sqrt{y}}{\sqrt{x}} \qquad ...(2)$$

$$\left(\frac{dy}{dx}\right)_{\left(\frac{1}{4},\frac{1}{4}\right)} = -\frac{\sqrt{\frac{1}{4}}}{\sqrt{\frac{1}{4}}} = -1$$

Differentiating (2) with respect to x, we get

$$\frac{d^2y}{dx^2} = -\frac{\sqrt{x}\left(\frac{1}{2\sqrt{y}}\frac{dy}{dx}\right) - \sqrt{y}\left(\frac{1}{2\sqrt{x}}\right)}{x}$$

$$\left(\frac{d^2y}{dx^2}\right)_{\left(\frac{1}{4},\frac{1}{4}\right)} = -\frac{\sqrt{\frac{1}{4}}\left(\frac{1}{2\sqrt{\frac{1}{4}}}(-1)\right) - \sqrt{\frac{1}{4}}\left(\frac{1}{2\sqrt{\frac{1}{4}}}\right)}{\left(\frac{1}{4}\right)} = \frac{\frac{1}{2}+\frac{1}{2}}{\left(\frac{1}{4}\right)} = 4$$

We know that $\rho = \frac{(1+y_1^2)^{3/2}}{y_2} = \frac{(1+1)^{3/2}}{4} = \frac{2^{3/2}}{4} = \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}$

Example 2: Find the radius of curvature at the point (c,c) on the curve $xy = c^2$

(Nov-2001)

Solution:

Given that $xy = c^2$

$$y = \frac{c^2}{x} \qquad \text{...(1)}$$

Differentiating (1) with respect to x, we get,

$$y_1 = \frac{dy}{dx} = c^2\left(-\frac{1}{x^2}\right) = \frac{-c^2}{x^2}$$

$$y_2 = \frac{d^2y}{dx^2} = -c^2\left(-\frac{2}{x^3}\right) = \frac{2c^2}{x^3}$$

$$y_1 = \left(\frac{dy}{dx}\right)_{(c,c)} = \frac{-c^2}{c^2} = -1$$

$$y_2 = \left(\frac{d^2y}{dx^2}\right)_{(c,c)} = \frac{2c^2}{c^3} = \frac{2}{c}$$

We know that, the radius of curvature is $\rho = \dfrac{(1+y_1^2)^{3/2}}{y_2}$

$$= \frac{\left[1+(-1)^2\right]^{3/2}}{\left(\frac{2}{c}\right)} = \frac{(2)^{3/2}}{\left(\frac{2}{c}\right)} = 2\sqrt{2} \times \frac{c}{2}$$

Radius of curvature $\rho = c\sqrt{2}$

Example 3: Find the radius of curvature at the point $(a,0)$ on the curve

$y^2 = \dfrac{a^2(a-x)}{x}$ **(Nov-2004)**

Solution:

Given that $y^2 = \dfrac{a^2(a-x)}{x}$

$$xy^2 = a^3 - a^2x \quad \text{...(1)}$$

Differentiating (1) with respect to x,

$$x(2y)\frac{dy}{dx} + y^2 = -a^2$$

$$2xy\frac{dy}{dx} = -a^2 - y^2$$

$$\frac{dy}{dx} = \frac{-a^2 - y^2}{2xy} \quad \text{...(2)}$$

$$\left(\frac{dy}{dx}\right)_{(a,0)} = \frac{-a^2}{0} = \infty$$

$\therefore$ we use the formula $\rho = \dfrac{\left\{1+\left(\dfrac{dx}{dy}\right)^2\right\}^{3/2}}{\left(\dfrac{d^2x}{dy^2}\right)}$ ...(3)

From (2) we get, $\dfrac{dx}{dy} = \dfrac{2xy}{-a^2 - y^2} = \dfrac{2xy}{-(a^2+y^2)}$ ...(4)

$$y_1 = \left(\frac{dx}{dy}\right)_{(a,0)} = \frac{2a(0)}{-(a^2+0)} = 0$$

Differentiating (4) with respect to x, we get,

$$\frac{d^2x}{dy^2} = -\frac{(y^2+a^2)\left[2x+2y\dfrac{dx}{dy}\right] - 2xy(2y)}{(y^2+a^2)^2}$$

$$\left(\frac{d^2x}{dy^2}\right)_{(a,0)} = -\frac{(0+a^2)\left[2a+2y(0)\right]-2a(0)}{(0+a^2)^2} = -\frac{2a^3}{a^4}$$

$$y_2 = \frac{d^2x}{dy^2} = -\frac{2}{a}$$

The radius of curvature $\rho = \dfrac{\left\{1+\left(\dfrac{dx}{dy}\right)^2\right\}^{3/2}}{\left(\dfrac{d^2x}{dy^2}\right)} = \dfrac{(1+0)^{3/2}}{\left(\dfrac{-2}{a}\right)} = -\dfrac{a}{2}$

$\rho = \dfrac{a}{2}$ (omit negative sign)

Example 4: Show that the radius of curvature at any point (x, y) of the catenary $y = c\cosh\frac{x}{c}$ is $\frac{y^2}{c}$ and equal to the length of the portion of the normal intercepted between the curve and the axis of x.

Solution:

Given that $y = c\cosh\frac{x}{c}$ …(1)

Differentiating (1) with respect to x, we get

$$y_1 = \frac{dy}{dx} = c\sinh\left(\frac{x}{c}\right)\frac{1}{c} = \sinh\left(\frac{x}{c}\right)$$

Differentiating (2) with respect to x, we get

$$y_2 = \frac{d^2y}{dx^2} = \frac{1}{c}\cosh\left(\frac{x}{c}\right)$$

Radius of Curvature $\rho = \frac{\left[1+y_1^2\right]^{3/2}}{y_2}$

$$= \frac{\left[1+\sinh^2\left(\frac{x}{c}\right)\right]^{3/2}}{\frac{1}{c}\cosh\left(\frac{x}{c}\right)} \qquad \left[\cosh^2 x - \sinh^2 x = 1\right]$$

$$= \frac{\left[\cosh^2\left(\frac{x}{c}\right)\right]^{3/2}}{\frac{1}{c}\cosh\left(\frac{x}{c}\right)} = \frac{c\cosh^3\left(\frac{x}{c}\right)}{\cosh\left(\frac{x}{c}\right)} = c\cosh^2\left(\frac{x}{c}\right) = c\frac{y^2}{c^2}$$

Radius of Curvature $\rho = \frac{y^2}{c}$

The length of the normal at any point (x,y) = $y\sqrt{1+\left(\frac{dy}{dx}\right)^2}$

$$= c\cosh\left(\frac{x}{c}\right)\sqrt{1+\sinh^2\left(\frac{x}{c}\right)}$$

$$= c\cosh^2\left(\frac{x}{c}\right) = \rho$$

Hence proved.

Example 5: Find the radius of curvature of $y = \log(\sec x)$ at any point on it.

Solution:

Given that $y = \log(\sec x)$

Differentiating with respect to x, we get $\frac{dy}{dx} = \frac{1}{\sec x}\sec x \tan x$

$$y_1 = \frac{dy}{dx} = \tan x$$

$$y_2 = \frac{d^2y}{dx^2} = \sec^2 x$$

Radius of curvature $\rho = \frac{\left[1+y_1^2\right]^{3/2}}{y_2}$

$$= \frac{\left[1+\tan^2 x\right]^{3/2}}{\sec^2 x} = \frac{\sec^3 x}{\sec^2 x}$$

Radius of Curvature $\rho = \sec x$

Example 6: Find the point on the parabola $y^2 = 4x$ at which the radius of curvature is $4\sqrt{2}$

Solution:

Let (x_0, y_0) be the point on $y^2 = 4x$ at which the radius of curvature is $4\sqrt{2}$.

Given that $y^2 = 4x$ ...(1)

Differentiate with respect to x,

$$2y\frac{dy}{dx} = 4$$

$$\frac{dy}{dx} = \frac{4}{2y} = \frac{2}{y} \qquad ...(2)$$

$$y_1 = \left(\frac{dy}{dx}\right)_{(x_0, y_0)} = \frac{2}{y_0}$$

Differentiating (2) with respect to 'x', we get

$$\frac{d^2y}{dx^2} = -\frac{2}{y^2}\frac{dy}{dx} = -\frac{2}{y^2}\left(\frac{2}{y}\right) = -\frac{4}{y^3}$$

$$y_2 = \left(\frac{d^2y}{dx^2}\right)_{(x_0, y_0)} = -\frac{4}{y_0^3}$$

Radius of curvature $\rho = \dfrac{\left[1 + y_1^2\right]^{3/2}}{y_2}$

Given that $\rho = 4\sqrt{2}$

Therefore, $4\sqrt{2} = \dfrac{\left[1 + \left(\dfrac{2}{y_0}\right)^2\right]^{3/2}}{\left(-\dfrac{4}{y_0^3}\right)} = -\left[\dfrac{y_0^2 + 4}{y_0^2}\right]^{3/2} \times \dfrac{y_0^3}{4}$

$$4\sqrt{2} = -\frac{\left(4 + y_0^2\right)^{3/2}}{y_0^3}\frac{y_0^3}{4}$$

$$16\sqrt{2} = -\left(4 + y_0^2\right)^{3/2}$$

$$-16\sqrt{2}=\left(4+y_0^2\right)^{3/2}$$

Squaring on both sides, we get

$$512=\left(4+y_0^2\right)^3 \qquad (y^2=4x \Rightarrow y_0^2=4x_0)$$

$$=(4+4x_0)^3$$

$$512=4^3(1+x_0)^3$$

$$(1+x_0)^3=\frac{512}{64}$$

$$(1+x_0)^3=8$$

$$(1+x_0)^3=2^3$$

$$1+x_0=2\text{ , } x_0=1$$

$$y_0^2=4x_0=4$$

$$y_0=\pm 2$$

$\therefore$ The required points are $(1,2)$ and $(1,-2)$ on the parabola $y^2=4x$ at which the radius of curvature is $4\sqrt{2}$

Example 7: Find the radius of curvature for the curve $x^3+y^3=3axy$ at $\left(\frac{3a}{2},\frac{3a}{2}\right)$

Solution:

Given that $x^3+y^3=3axy$

...(1)

Differentiating (1) with respect to x, we get

$$3x^2+3y^2\frac{dy}{dx}=3a\left[x\frac{dy}{dx}+y\right]$$

$$\frac{dy}{dx}(3y^2-3ax)=3ay-3x^2$$

$$\frac{dy}{dx}=\frac{ay-x^2}{y^2-ax}$$

$$...(2)$$

$$\left(\frac{dy}{dx}\right)_{\left(\frac{3a}{2},\frac{3a}{2}\right)}=\frac{a.\frac{3a}{2}-\frac{9a^2}{4}}{\frac{9a^2}{4}-a\frac{3a}{2}}=\frac{-3a^2}{3a^2}$$

$$y_1=-1$$

Differentiating (2) with respect to x, we get

$$\frac{d^2y}{dx^2}=\frac{(y^2-ax)\left[a\frac{dy}{dx}-2x\right]-\left(ay-x^2\right)\left[2y\frac{dy}{dx}-a\right]}{(y^2-ax)^2}$$

$$\left(\frac{dy}{dx}\right)_{\left(\frac{3a}{2},\frac{3a}{2}\right)}=\frac{\left(\frac{9a^2}{4}-a\frac{3a}{2}\right)\left[a(-1)-2\left(\frac{3a}{2}\right)\right]-\left(\frac{a\left(3a\right)}{2}-\frac{9a^2}{4}\right)\left[2\left(\frac{3a}{2}\right)(-1)-a\right]}{\left(\frac{9a^2}{4}-\frac{a3a}{2}\right)^2}$$

$$=\frac{\left(\frac{3a^2}{4}\right)\left(\frac{-8a}{2}\right)-\left(\frac{-3a^2}{4}\right)\left(\frac{-8a}{2}\right)}{\frac{9a^4}{16}}=\frac{-3a^3-3a^3}{\frac{9a^4}{16}}$$

$$y_2=\left(\frac{d^2y}{dx^2}\right)_{\left(\frac{3a}{2},\frac{3a}{2}\right)}=-\frac{32}{3a}$$

Radius of curvature $\rho=\frac{\left[1+y_1^2\right]^{3/2}}{y_2}$

$$= \frac{\left[1+(-1)^2\right]^{3/2}}{\frac{-32}{3a}} = -2^{3/2} \times \frac{3a}{32} = -\frac{3a \times 2 \times \sqrt{2}}{32}$$

Radius of curvature $\rho = -\frac{3a\sqrt{2}}{16}$

Example 8: Show that ρ at any point $(a\cos^3\theta, a\sin^3\theta)$ on the curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ is $3asin\theta\cos\theta$ **(May-2000, Nov-2003, 2011)**

Solution:

Given that $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$

The parametric equations of the curve are $x = a\cos^3\theta, \quad y = a\sin^3\theta$

$$\frac{dx}{d\theta} = -3a\cos^2\theta\sin\theta, \quad \frac{dy}{d\theta} = 3a\sin^2\theta\cos\theta$$

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = -\frac{3a\sin^2\theta\cos\theta}{3a\cos^2\theta\sin\theta}$$

$$y_1 = -\tan\theta$$

$$\frac{d^2y}{dx^2} = \frac{d}{d\theta}\left(\frac{dy}{dx}\right)\frac{d\theta}{dx}$$

$$= \frac{d}{d\theta}(-\tan\theta)\frac{d\theta}{dx}$$

$$= -\sec^2\theta\frac{d\theta}{dx} = -\sec^2\theta\left(\frac{1}{-3a\cos^2\theta\sin\theta}\right)$$

$$y_2 = \frac{d^2y}{dx^2} = \frac{1}{3a\cos^4\theta\sin\theta}$$

Radius of curvature $\rho = \dfrac{\left[1+y_1^2\right]^{3/2}}{y_2}$

$$\rho = \frac{\left[1+\tan^2\theta\right]^{3/2}}{\left(\dfrac{1}{3a\cos^4\theta\sin\theta}\right)} = \sec^3\theta 3a\cos^4\theta\sin\theta$$

Radius of curvature $\rho = 3a\sin\theta\cos\theta$

Example 9: Find the radius of curvature of the point θ on the curve

$x = a(\theta+\sin\theta), \quad y = a(1-\cos\theta)$

Solution:

Given that $x = a(\theta+\sin\theta), \quad y = a(1-\cos\theta)$

$$\frac{dx}{d\theta} = a(1+\cos\theta), \qquad \frac{dy}{d\theta} = a\sin\theta$$

$$\frac{dy}{dx} = \frac{\dfrac{dy}{d\theta}}{\dfrac{dx}{d\theta}} = \frac{a\sin\theta}{a(1+\cos\theta)} \qquad \left[\begin{array}{l} \because \sin 2\theta = 2\sin\theta\cos\theta \\ \cos^2\theta = \dfrac{1+\cos 2\theta}{2} \end{array}\right]$$

$$= \frac{2\sin\left(\dfrac{\theta}{2}\right)\cos\left(\dfrac{\theta}{2}\right)}{2\cos^2\left(\dfrac{\theta}{2}\right)} = \frac{\sin\left(\dfrac{\theta}{2}\right)}{\cos\left(\dfrac{\theta}{2}\right)}$$

$$\frac{dy}{dx} = \tan\left(\frac{\theta}{2}\right)$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$$

$$= \frac{d}{d\theta}\left(\frac{dy}{dx}\right)\frac{d\theta}{dx}$$

$$= \frac{d}{d\theta}\tan\left(\frac{\theta}{2}\right)\frac{1}{a(1+\cos\theta)} = \frac{1}{2}\sec^2\left(\frac{\theta}{2}\right)\frac{1}{2a\cos^2\left(\frac{\theta}{2}\right)}$$

$$= \frac{1}{4a\cos^4\left(\frac{\theta}{2}\right)}$$

Radius of curvature $\rho = \frac{\left[1+y_1^2\right]^{3/2}}{y_2}$

$$= \frac{\left[1+\tan^2\left(\frac{\theta}{2}\right)\right]^{3/2}}{\left(\frac{1}{4a\cos^4\left(\frac{\theta}{2}\right)}\right)} = \left(\sec^2\left(\frac{\theta}{2}\right)\right)^{3/2} \times 4a\cos^4\left(\frac{\theta}{2}\right)$$

$$= 4a\sec^3\left(\frac{\theta}{2}\right)\cos^4\left(\frac{\theta}{2}\right)$$

Radius of curvature $\rho = 4a\cos\left(\frac{\theta}{2}\right)$

Example 10: Find the radius of curvature at the point 't' on the curve

$x = a(\cos t + t\sin t), \quad y = a(\sin t - t\cos t)$

Solution:

Given that $x = a(\cos t + t\sin t), \;\; y = a(\sin t - t\cos t)$

$$\frac{dx}{dt} = a(-\sin t + t\cos t + \sin t) = at\cos t$$

$$\frac{dy}{dt} = a(\cos t + t\sin t - \cos t) = at\sin t$$

$$\frac{dy}{dx}=\frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)}=\frac{at\sin t}{at\cos t}$$

$$y_1=\tan t$$

$$\frac{d^2y}{dx^2}=\frac{d}{dt}\left(\frac{dy}{dx}\right)\frac{dt}{dx}$$

$$=\frac{d}{dt}(\tan t)\frac{1}{at\cos t} \quad =\sec^2 t\times\frac{1}{at\cos t}$$

$$y_2=\frac{1}{at}\sec^3 t$$

Radius of curvature $\rho=\frac{\left[1+y_1^2\right]^{3/2}}{y_2}$

$$=\frac{\left[1+\tan^2 t\right]^{3/2}}{\left(\frac{\sec^3 t}{at}\right)}=\left(\sec^2 t\right)^{3/2}\times\frac{at}{\sec^3 t}$$

Radius of curvature $\rho=at$

Example 11: Find the radius of curvature at the point θ on the curve $y=a\sec\theta$ $x=a\log(\sec\theta+\tan\theta)$,

Solution:

Given that $x=a\log(\sec\theta+\tan\theta)$, $\quad y=a\sec\theta$

$$\frac{dx}{d\theta}=\frac{a}{\sec\theta+\tan\theta}(\sec\theta\tan\theta+\sec^2\theta)$$

$$=a\frac{\sec\theta(\tan\theta+\sec\theta)}{(\tan\theta+\sec\theta)}=a\sec\theta$$

$$\frac{dy}{d\theta} = a\sec\theta\tan\theta$$

$$\frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} = \frac{a\sec\theta\tan\theta}{a\sec\theta}$$

$$y_1 = \tan\theta$$

$$\frac{d^2y}{dx^2} = \frac{d}{d\theta}\left(\frac{dy}{dx}\right)\frac{d\theta}{dx}$$

$$= \frac{d}{d\theta}\left(\tan\theta\right)\frac{1}{a\sec\theta} = \frac{\sec^2\theta}{a\sec\theta}$$

$$y_2 = \frac{\sec\theta}{a}$$

Radius of curvature $\rho = \frac{\left[1+y_1^2\right]^{3/2}}{y_2}$

$$= \frac{(1+\tan^2\theta)^{3/2}}{\left(\frac{\sec\theta}{a}\right)} = \sec^3\theta \times \frac{a}{\sec\theta} = a\sec^2\theta$$

Radius of curvature $\rho = a\sec^2\theta$

Example 12: For the cardioid $r = a(1+\cos\theta)$ prove that $\frac{\rho^2}{r}$ constant

Solution:

Given that $r = a(1+\cos\theta)$

$$\frac{dr}{d\theta} = -a\sin\theta$$

$$\frac{d^2r}{d\theta^2} = -a\cos\theta$$

Radius of curvature in polar coordinate $\rho = \dfrac{\left[r^2 + \left(\dfrac{dr}{d\theta}\right)^2\right]^{3/2}}{r^2 + 2\left(\dfrac{dr}{d\theta}\right)^2 - r\dfrac{d^2r}{d\theta^2}}$ ...(1)

Consider $\left[r^2 + \left(\dfrac{dr}{d\theta}\right)^2\right]^{\frac{3}{2}} = \left[a^2(1+\cos\theta)^2 + a^2\sin^2\theta\right]^{\frac{3}{2}}$

$$= \left[a^2 + a^2\cos^2\theta + 2a^2\cos\theta + a^2\sin^2\theta\right]^{\frac{3}{2}}$$

$$= \left[a^2 + a^2(\cos^2 + \sin^2\theta) + 2a^2\cos\theta\right]^{\frac{3}{2}}$$

$$= (a^2 + a^2 + 2a^2\cos\theta)^{\frac{3}{2}}$$

$$= (2a^2 + 2a^2\cos\theta)^{\frac{3}{2}}$$

$$= (2a^2)^{\frac{3}{2}}(1+\cos\theta)^{\frac{3}{2}}$$

$$= 2^{\frac{3}{2}}a^3(1+\cos\theta)^{\frac{3}{2}}$$ ...(2)

Consider $r^2 + 2\left(\dfrac{dr}{d\theta}\right)^2 - r\dfrac{d^2r}{d\theta^2} = a^2(1+\cos\theta)^2 + 2a^2\sin^2\theta - a(1+\cos\theta)(-a\cos\theta)$

$$= a^2 + a^2\cos^2\theta + 2a^2\cos\theta + 2a^2\sin^2\theta + a^2\cos\theta + a^2\cos^2\theta$$

$$= a^2 + 2a^2\cos^2\theta + 2a^2\sin^2\theta + 3a^2\cos\theta$$

$$= a^2 + 2a^2(\cos^2\theta + \sin^2\theta) + 3a^2\cos\theta$$

$$= a^2 + 2a^2 + 3a^2\cos\theta$$

$$= 3a^2 + 3a^2 \cos\theta$$

$$= 3a^2(1+\cos\theta) \qquad ...(3)$$

Substituting (2) and (3) in (1) we get

$$\rho = \frac{(a^2)^{3/2}(2)^{3/2}(1+\cos\theta)^{3/2}}{3a^2(1+\cos\theta)}$$

$$= \frac{2\sqrt{2}a^3(1+\cos\theta)(1+\cos\theta)^{1/2}}{3a^2(1+\cos\theta)}$$

$$= \frac{2\sqrt{2}}{3}\sqrt{a}\left[a(1+\cos\theta)\right]^{1/2}$$

$$= \frac{2}{3}\sqrt{2a}\sqrt{a(1+\cos\theta)}$$

$$\rho^2 = \frac{8a}{9}r$$

$$\frac{\rho^2}{r} = \frac{8a}{9}, \text{ a constant}$$

Hence proved.

Example 13: Show that the radius of curvature of the curve $r^n = a^n \cos n\theta$ **is** $\dfrac{a^n r^{-n+1}}{n+1}$

Hence prove that the radius of curvature of the curve $r^2 = a^2\cos^2\theta$ **is** $\dfrac{a^2}{3n}$**.**

Solution:

Given that $r^n = a^n \cos n\theta$ ...(1)

Take log on both sides of (1),

$$\log r^n = \log(a^n \cos n\theta)$$

$$n\log r = \log a^n + \log\cos n\theta$$

$$n\log r = n\log a + \log\cos n\theta$$

Differentiating with respect to θ, we get

$$\frac{n}{r}\frac{dr}{d\theta} = -\frac{n\sin n\theta}{\cos n\theta}$$

$$\frac{dr}{d\theta} = -r\tan n\theta \qquad ...(2)$$

$$\frac{d^2r}{d\theta^2} = -\left[rn\sec^2 n\theta + \tan n\theta\frac{dr}{d\theta}\right]$$

$$= -\left[rn\sec^2 n\theta - r\tan^2 n\theta\right]$$

$$= r\tan^2 n\theta - rn\sec^2 n\theta$$

Radius of curvature $\rho = \dfrac{\left[r^2 + \left(\dfrac{dr}{d\theta}\right)^2\right]^{3/2}}{r^2 + 2\left(\dfrac{dr}{d\theta}\right)^2 - r\dfrac{d^2r}{d\theta^2}}$

$$= \frac{(r^2 + r^2\tan^2 n\theta)^{3/2}}{r^2 + 2r^2\tan^2 n\theta - r(r\tan^2 n\theta - rn\sec^2 n\theta)}$$

$$= \frac{(r^2\sec^2 n\theta)^{3/2}}{r^2 n\sec^2 n\theta + r^2 + r^2\tan^2 n\theta}$$

$$= \frac{r^3\sec^3 n\theta}{r^2\sec^2 n\theta(n+1)} = \frac{r\sec n\theta}{(n+1)}$$

$$= \frac{r}{(n+1)\cos n\theta} = \frac{r}{(n+1)\dfrac{r^n}{a^n}}$$

Radius of curvature $\rho = \dfrac{a^n r^{-n+1}}{n+1}$ ...(3)

Put $n = 2$ the equation (1) becomes,

$$r^2 = a^2\cos^2\theta$$

From (4) $\rho = \dfrac{a^2 r^{-1}}{3} = \dfrac{a^2}{3r}$

Example 14: Find the radius of curvature at $(-2,0)$ on $y^2 = x^3 + 8$ (May 2011)

Solution:

Given that $y^2 = x^3 + 8$ ...(1)

$$2y\frac{dy}{dx} = 3x^2$$

$$\frac{dy}{dx} = \frac{3x^2}{2y} \qquad ...(2)$$

$$\left(\frac{dy}{dx}\right)_{(-2,0)} = \frac{12}{0} = \infty$$

$\therefore$ We use the formula $\rho = \dfrac{\left[1+y_1^2\right]^{3/2}}{y_2}$ where $y_1 = \dfrac{dx}{dy}$, $y_2 = \dfrac{d^2x}{dy^2}$

From (2), $\dfrac{dx}{dy} = \dfrac{2y}{3x^2}$ ...(3)

$$y_1 = \left(\frac{dx}{dy}\right)_{(-2,0)} = 0$$

Differentiating (3) with respect to y, we get

$$\frac{d^2x}{dy^2} = \frac{(3x^2)(2) - (2y)6x\dfrac{dx}{dy}}{9x^4}$$

$$y_2 = \left(\frac{d^2x}{dy^2}\right)_{(-2,0)} = \frac{24-0}{144} = \frac{1}{6}$$

Radius of curvature $\rho = \dfrac{[1+0]^{3/2}}{\left(\dfrac{1}{6}\right)} = 6$

Example 15: Find the radius of curvature for $x^4 + y^4 = 2$ **at** $(1,1)$ **(Jan- 2010)**

Solution:

Given that $x^4 + y^4 = 2$ …(1)

Diff (1) with respect to x,

$$4x^3 + 4y^3 \frac{dy}{dx} = 0 \quad ...(2)$$

$$\frac{dy}{dx} = \frac{-4x^3}{4y^3} \Rightarrow \left(\frac{dy}{dx}\right)_{(1,1)} = -1$$

Diff (2) with respect to x,

$$12x^2 + 12y^2\left(\frac{dy}{dx}\right)^2 + 4y^3\frac{d^2y}{dx^2} = 0$$

$$4y^3\frac{d^2y}{dx^2} = -12x^2 - 12y^2\left(\frac{dy}{dx}\right)^2$$

$$\frac{d^2y}{dx^2} = \frac{-12x^2 - 12y^2\left(\dfrac{dy}{dx}\right)^2}{4y^3}$$

$$\left(\frac{d^2y}{dx^2}\right)_{(1,1)} = \frac{-12-12(-1)^2}{4} = \frac{-24}{4} = -6$$

Radius of curvature $\rho = \dfrac{\left[1+y_1^2\right]^{3/2}}{y_2}$

$$= \frac{\left[1+(-1)^2\right]^{3/2}}{-6} = \frac{2^{3/2}}{-6} = \frac{2\sqrt{2}}{6} \text{ (Neglecting the negative sign)}$$

Exercise

1. Find the radius of curvature for $x^4 + y^4 = 2$ at $(1,1)$

2. Find the radius of curvature for $y = \dfrac{\log x}{x}$ at $x = 1$

3. Find the radius of curvature for $xy^2 = a^3 - x^3$ at $(a, 0)$

4. Find the radius of curvature at $\theta = \dfrac{\pi}{3}$ on $x = a\sin\theta,\ \ y = b\cos 2\theta$

5. Find the radius of curvature at t, $x = ct,\quad y = \dfrac{c}{t}$

6. Find the radius of the radius for $x = a\log\sec\theta,\quad y = a(\tan\theta - \theta)$

7. Find the radius of the curvature for the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ at $(a\cos\theta, b\sin\theta)$

8. Find the radius of curvature for $x^3 + y^3 = 2a^3$ at (a, a)

9. Find the radius of curvature for $r = a(1 - \cos\theta)$

10. Find the radius of curvature for e^x at $(0,1)$

11. Find the radius of curvature for $x = at^2,\quad y = 2at$

12. Find the radius of curvature for $x = a(t + \sin t),\quad y = a(1 - \cos t)$

Answers

1. $\rho = \dfrac{\sqrt{2}}{3}$

2. $\dfrac{2\sqrt{2}}{3}$

3. $\dfrac{3a}{2}$

4. $\dfrac{(a^2 + 12b^2)^{3/2}}{4ab}$

5. $\dfrac{c}{2t^3}(t^4 + 1)^{3/2}$

6. $a\tan\theta\sec\theta$

7. $\dfrac{(a^2\sin^2\theta + b^2\cos^2\theta)^{3/2}}{ab}$

8. $\dfrac{a}{\sqrt{2}}$

9. $\dfrac{2}{3}\sqrt{2ar}$

10. $2\sqrt{2}$

11. $\dfrac{\left(1 + t^2\right)^{\frac{3}{2}}}{t}$

12. $4a\cos\left(\dfrac{t}{2}\right)$

1.5. Centre of curvature - coordinates

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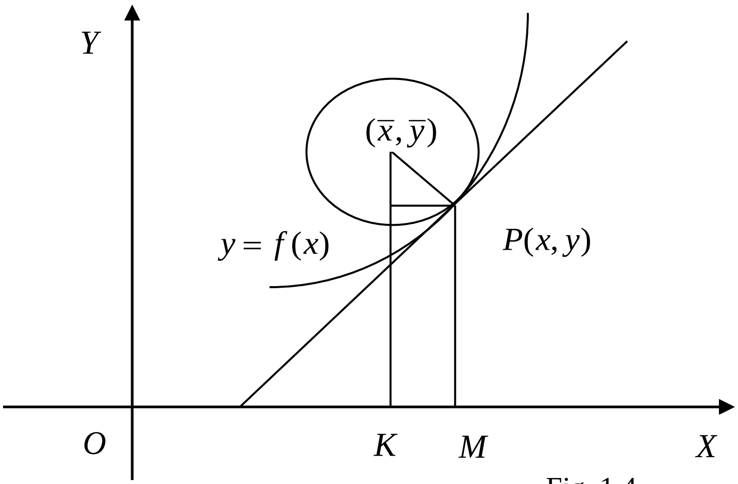
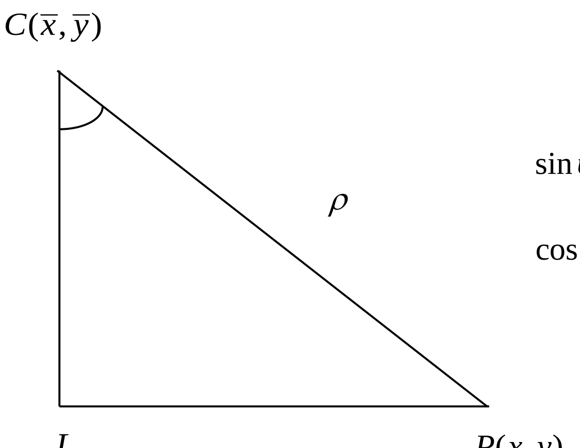


Fig. 1.4

Since ρ is radius of curvature, $CP = \rho$. Let ψ be the angle made by the tangent at P with OX. Draw CK, PM $\perp$ OX and PL perpendicular to CK.



$$\sin\psi = \frac{LP}{CP}$$

$$\cos\phi = \frac{LC}{CP}$$

Fig. 1.5.

Then $\angle PCL = \psi$

$$\bar{x} = OK = OM - KM = OM - LP$$

$$= OM - CP.\sin\psi$$

$$= x - \rho\sin\psi \qquad \text{...(1)}$$

$$\bar{y} = KC = KL + LC$$

$$= MP + CP\cos\psi$$

$$= y + \rho\cos\psi \qquad \text{...(2)}$$

We know that, the radius of curvature, $\rho = \dfrac{\left[1+\left(\dfrac{dy}{dx}\right)^2\right]^{3/2}}{\left(\dfrac{d^2y}{dx^2}\right)} = \dfrac{\left[1+y_1^2\right]^{3/2}}{y_2}$

$$\sin\psi = \frac{\sin\psi}{\cos\psi}\cos\psi = \frac{\tan\psi}{\sec\psi} = \frac{\tan\psi}{\sqrt{1+\tan^2\psi}}$$

$$= \frac{\dfrac{dy}{dx}}{\sqrt{1+\left(\dfrac{dy}{dx}\right)^2}} = \frac{y_1}{\sqrt{1+y_1^2}}$$

From (1), $\bar{x} = x - \rho\sin\psi = x - \dfrac{(1+y_1^2)^{3/2}}{y_2}\dfrac{y_1}{\sqrt{1+y_1{}^2}}$

$$= x - \frac{(1+y_1^2)^{3/2}}{y_2}\frac{y_1}{\sqrt{1+y_1^2}} = x - \frac{y_1(1+y_1^2)}{y_2}$$

From (2), $\bar{y} = y + \rho\cos\psi = y + \dfrac{(1+y_1^2)^{3/2}}{y_2}\dfrac{1}{\sqrt{1+y_1^2}}$,

$$\bar{y} = y + \frac{(1+y_1^2)}{y_2} \qquad \left[\begin{aligned} \because \cos\psi &= 1-\sin^2\psi \\ &= 1-\frac{y_1^2}{1+y_1^2} = \frac{1}{1+y_1^2} \end{aligned}\right]$$

1.6. Circle of curvature

The circle of curvature is the circle whose centre is the centre of curvature and radius is the radius of curvature.

$\therefore$ The equation of circle of curvature is, $(x-\bar{x})^2 + (y-\bar{y})^2 = \rho^2$

Important formulae:

Formula to find centre of curvature:-

$$\bar{x} = x - \frac{y_1(1+y_1^2)}{y_2} \; ; \; \bar{y} = y + \frac{(1+y_1^2)}{y_2}$$

Formula to find circle of curvature:-

The equation of circle of curvature is given by $(x-\bar{x})^2 + (y-\bar{y})^2 = \rho^2$

Example 1: Find the coordinates of centre of curvature on $y = x^3$ at $(-3,9)$

Solution:

Given that $y = x^3$

$$\frac{dy}{dx} = y_1 = 3x^2$$

$$\frac{d^2y}{dx^2} = y_2 = 6x$$

At $(-3,9)$, $y_1 = 3(-3)^2 = 27$

$$y_2 = 6(-3) = -18$$

$$\bar{x} = x - \frac{y_1}{y_2}(1 + y_1^2)$$

$$= -3 - \frac{27}{(-18)}(1+729) \quad = -3 + \frac{3}{2}(730)$$

$$= 1092$$

$$\bar{y} = y + \frac{(1+y_1^2)}{y_2}$$

$$= 9 + \frac{1+729}{-18} = 9 + \frac{730}{-18} = 9 - \frac{365}{9}$$

$$= \frac{81-365}{9} = \frac{-284}{9}$$

The centre of curvature is $\left(1092, \frac{-284}{9}\right)$

Example 2: Find the centre of curvature of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at (a,b)

Solution:

Given that $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ...(1)

Differentiating (1) with respect to *x*,

$$\frac{1}{a^2}(2x) + \frac{1}{b^2}(2y)\frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{\frac{2x}{a^2}}{\frac{2y}{b^2}} = -\frac{xb^2}{ya^2} \qquad ...(2)$$

$$\left(\frac{dy}{dx}\right)_{(a,b)} = -\frac{ab^2}{ba^2} = -\frac{b}{a}$$

$$y_1 = -\frac{b}{a}$$

Differentiating (2) with respect to x,

$$\frac{d^2y}{dx^2} = -\frac{b^2}{a^2}\left[\frac{y(1) - x\frac{dy}{dx}}{y^2}\right]$$

$$\left(\frac{d^2y}{dx^2}\right)_{(a,b)} = -\frac{b^2}{a^2}\left[\frac{b - a\left(-\frac{b}{a}\right)}{b^2}\right] = -\frac{b^2}{a^2}\left[\frac{2b}{b^2}\right]$$

$$y_2 = -\frac{2b}{a^2}$$

$$\therefore \bar{x} = x - \frac{y_1(1+y_1^2)}{y_2}$$

$$= a - \left[\frac{\left(-\frac{b}{a}\right)\left[1+\frac{b^2}{a^2}\right]}{\left(\frac{-2b}{a^2}\right)}\right] = a - \left[\frac{\frac{b}{a}\left[\frac{a^2+b^2}{a^2}\right]}{\frac{2b}{a^2}}\right]$$

$$= a - \left[\frac{b(a^2+b^2)a^2}{2ba^3}\right] = a - \left[\frac{a^2+b^2}{2a}\right]$$

$$= \frac{2a^2 - a^2 - b^2}{2a} = \frac{a^2 - b^2}{2a}$$

$$\bar{y} = y + \left[\frac{1+y_1^2}{y_2}\right]$$

$$= b + \left[\frac{1 + \frac{b^2}{a^2}}{\frac{-2b}{a^2}} \right] = b + \left[\frac{\frac{a^2 + b^2}{a^2}}{\frac{-2b}{a^2}} \right]$$

$$= b + \frac{a^2 + b^2}{a^2} \times \frac{a^2}{-2b} = b + \frac{a^2 + b^2}{-2b}$$

$$= \frac{-2b^2 + a^2 + b^2}{-2b} = \frac{a^2 - b^2}{-2b}$$

$$\bar{y} = \frac{b^2 - a^2}{2b}$$

Centre of curvature at (a,b) is, $\bar{x} = \dfrac{a^2 - b^2}{2a}$, $\bar{y} = \dfrac{b^2 - a^2}{2b}$

Example 3: Find the circle of curvature at $(0,0)$ on $x + y = x^2 + y^2 + x^3$ (Nov- 2011)

Solution:

Given that $x + y = x^2 + y^2 + x^3$

$$x^3 + x^2 + y^2 - x - y = 0 \qquad ...(1)$$

Differentiating (1) with respect to *x*,

$$3x^2 + 2x + 2y\frac{dy}{dx} - 1 - \frac{dy}{dx} = 0$$

$$(2y - 1)\frac{dy}{dx} = 1 - 3x^2 - 2x$$

$$\frac{dy}{dx} = \frac{1 - 3x^2 - 2x}{2y - 1} \qquad ...(2)$$

$$\left(\frac{dy}{dx}\right)_{(0,0)} = \frac{1}{-1} = -1$$

$$y_1 = -1$$

Differentiating (2) with respect to x,

$$\frac{d^2y}{dx^2} = \frac{(2y-1)(-6x-2)-(1-3x^2-2x)\left(2\frac{dy}{dx}\right)}{(2y-1)^2}$$

$$\left(\frac{d^2y}{dx^2}\right)_{(0,0)} = \frac{((-1)(-2))-(1)(-2)}{(-1)^2} = \frac{2+2}{1}$$

$$y_2 = 4$$

$$\bar{x} = x - \frac{y_1}{y_2}(1+y_1^2)$$

$$= 0 - \frac{(-1)}{4}(1+1) = \frac{1}{2}$$

$$\bar{y} = y + \frac{(1+y_1^2)}{y_2}$$

$$= 0 + \frac{(1+1)}{4} = \frac{1}{2}$$

Circle of curvature is given by $(x-\bar{x})^2 + (y-\bar{y})^2 = \rho^2$ ...(3)

Radius of curvature $\rho = \frac{\left[1+y_1^2\right]^{3/2}}{y_2} = \frac{\left[1+1\right]^{3/2}}{4} = \frac{(2)^{3/2}}{4} = \frac{2\sqrt{2}}{4}$

$$\rho = \frac{1}{\sqrt{2}}$$

Equation (3) becomes $(x-\frac{1}{2})^2 + (y-\frac{1}{2})^2 = \frac{1}{2}$

(i.e.) $x^2 - x + \frac{1}{4} + y^2 - y + \frac{1}{4} = \frac{1}{2}$

(i.e.) $x^2 + y^2 - x - y = 0$

Exercise

1. Find the centre of curvature of the curve $y = 2x^3 + 3x^2 - 4x + 7$ at the point $(-1,1)$

2. Find the centre of curvature for $y = x^2$ at $\left(\frac{1}{2}, -\frac{1}{2}\right)$

3. Find the centre of curvature of the curve $y = 3x^3 + 2x^2 - 3$ at $(0,-3)$

4. Find the centre of curvature of the curve $x = a(\cos t + t\sin t)$, $y = a(\sin t - t\cos t)$ at t

5. Find the circle of curvature for the curve $y^2 = 12x$ at $(3,6)$

6. Show that the circle of curvature of the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$ at the point $\left[\frac{a}{4}, \frac{a}{4}\right]$ is

$$\left(x - \frac{3a}{4}\right)^2 + \left(y - \frac{3a}{4}\right)^2 = \frac{1}{2}a^2$$

7. Find the equation of the circle of curvature at (c,c) on $xy = c^2$

8. Find the centre of curvature at the curve $y = 2x^3 + 3x^2 - 4x + 7$ at the point $(-1,1)$

Answer

1. $\left(\frac{-37}{6}, \frac{-11}{6}\right)$

2. $\left(-\frac{1}{2}, \frac{3}{4}\right)$

3. $\left(0, \frac{-11}{4}\right)$

4. $(a\cos t, a\sin t)$

5. $x^2 y^2 - 30x + 12y - 27 = 0$

7. $(x-2c)^2 + (y-2c)^2 = 2c^2$

8. $\left(\frac{-37}{3}, \frac{-11}{6}\right)$

1.7. Evolutes

The locus of center of curvature$\left(\bar{x}, \bar{y}\right)$, is the evolutes of the given curve and the given curve is called the involutes.

Let the given curve be $y = f(x)$

The coordinates of centre of curvature are

$$\bar{x} = x - \frac{y_1\left(1+y_1^2\right)}{y_2} \quad \text{and} \quad \bar{y} = y + \frac{\left(1+y_1^2\right)}{y_2}$$

Where $y_1 = \dfrac{dy}{dx}$ and $y_2 = \dfrac{d^2y}{dx^2}$

Some standard parametric transformations

Sl. No.	Curve	Parametric form
1.	Circle $x^2 + y^2 = a^2$	$x = a\cos\theta, \quad y = a\sin\theta$
2.	Parabola $y^2 = 4ax$	$x = at^2, \quad y = 2at$
3.	Parabola $x^2 = 4ay$	$x = 2at, \quad y = at^2$
4.	Ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$	$x = a\cos\theta, \quad y = b\sin\theta$
5.	Hyperbola $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$	$x = a\sec\theta, \quad y = b\tan\theta$
6.	Rectangular Hyperbola $xy = c^2$	$x = ct, y = \dfrac{c}{t}$
7.	Astroid $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$	$x = a\cos^3\theta, y = a\sin^3\theta$

Example 1: Find the equation of evolute of the parabola $y^2 = 4ax$

(Jan- 2010, May- 2011,12)

Solution:

Given that $y^2 = 4ax$

The parametric equations of the parabola $y^2 = 4ax$ are $x = at^2, \quad y = 2at$

$$\frac{dx}{dt} = 2at \qquad \frac{dy}{dt} = 2a$$

$$\therefore \frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)} = \frac{2a}{2at}$$

$$y_1 = \frac{1}{t}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dt}\right) = \frac{d}{dx}\left(\frac{1}{t}\right)$$

$$= -\frac{1}{t^2}\frac{dt}{dx} = -\frac{1}{t^2}\frac{1}{2at}$$

$$y_2 = -\frac{1}{2at^3}$$

$$\bar{x} = x - \frac{y_1(1+y_1^2)}{y_2}$$

$$= at^2 - \frac{\frac{1}{t}\left(1+\frac{1}{t^2}\right)}{-\frac{1}{2at^3}} = at^2 + \frac{2at^3}{t}\left(1+\frac{1}{t^2}\right)$$

$$= at^2 + \frac{2at^3}{t}\left(\frac{t^2+1}{t^2}\right) = at^2 + 2a(t^2+1)$$

$$= at^2 + 2at^2 + 2a$$

$$\bar{x} = 3at^2 + 2a \qquad \text{...(1)}$$

$$\bar{y} = y + \frac{(1+y_1^2)}{y_2}$$

$$= 2at + \frac{\left(1+\frac{1}{t^2}\right)}{\left(-\frac{1}{2at^3}\right)} = 2at - 2at^3\left(1+\frac{1}{t^2}\right)$$

$$= 2at - 2at^3\left(\frac{t^2+1}{t^2}\right) = 2at - 2at(t^2+1)$$

$$= 2at - 2at^3 - 2at$$

$$\bar{y} = -2at^3 \qquad ...(2)$$

From (1), $3at^2 = \bar{x} - 2a$

$$t^2 = \frac{\bar{x}-2a}{3a}$$

$$t = \sqrt{\frac{\bar{x}-2a}{3a}}$$

Substituting ‘t’ in (2)

$$\bar{y} = -2a\left[\sqrt{\frac{\bar{x}-2a}{3a}}\right]^3$$

$$= -2a\left[\left(\frac{\bar{x}-2a}{3a}\right)^{1/2}\right]^3 = -2a\left[\frac{\bar{x}-2a}{3a}\right]^{3/2}$$

Squaring on both sides,

$$\bar{y}^2 = 4a^2\left[\frac{\bar{x}-2a}{3a}\right]^3 = 4a^2\,\frac{(\bar{x}-2a)^3}{27a^3}$$

$$= \frac{4(\bar{x}-2a)^3}{27a}$$

$$27a\bar{y}^2 = 4(\bar{x}-2a)^3$$

Replace $\bar{x}$ by x and $\bar{y}$ by y , we get $27ay^2 = 4(x-2a)^3$

This is the equation of evolute of the parabola $y^2 = 4ax$

Example 2: Find the equation of evolute $x^2 = 4ay$ **(April -2010)**

Solution:

Given that $x^2 = 4ay$

The parametric coordinates of the parabola $x^2 = 4ay$ are $x = 2at$, $y = at^2$

$$\frac{dx}{dt} = 2a \qquad \frac{dy}{dt} = 2at$$

$$\frac{dy}{dt} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2at}{2a}$$

$$y_1 = t$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$$

$$= \frac{d}{dt}\left(\frac{dy}{dx}\right)\frac{dt}{dx} = \frac{d}{dt}(t)\frac{dt}{dx}$$

$$= 1 \cdot \frac{dt}{dx}$$

$$y_2 = \frac{1}{2a}$$

$$\bar{x} = x - \frac{y_1(1+y_1^2)}{y_2}$$

$$= 2at - \frac{t(1+t^2)}{\left(\frac{1}{2a}\right)} = 2at - 2at(1+t^2)$$

$$= 2at - 2at - 2at^3$$

$$\bar{x} = -2at^3 \qquad \ldots(1)$$

$$\bar{y} = y + \frac{(1+y_1^2)}{y_2}$$

$$= at^2 + \frac{(1+t^2)}{\left(\frac{1}{2a}\right)} = at^2 + 2a(1+t^2)$$

$$= at^2 + 2a + 2at^2$$

$$\bar{y} = 3at^2 + 2a \qquad \ldots(2)$$

From (2), $3at^2 = \bar{y} - 2a$

$$t^2 = \frac{\bar{y} - 2a}{3a}$$

$$t = \sqrt{\frac{\bar{y} - 2a}{3a}}$$

Substituting 't' in (1), we get $\bar{x} = -2a\left[\sqrt{\frac{\bar{y} - 2a}{3a}}\right]^3$

$$= -2a\left[\left(\frac{\bar{y} - 2a}{3a}\right)^{1/2}\right]^3 = -2a\left[\frac{\bar{y} - 2a}{3a}\right]^{3/2}$$

Squaring both sides, we get

$$\bar{x}^2 = 4a^2\left(\frac{\bar{y} - 2a}{3a}\right)^3$$

$$= \frac{4a^2(\bar{y} - 2a)^3}{27a^3}$$

$$\bar{x}^2 = \frac{4(\bar{y} - 2a)^3}{27a}$$

$$27a\bar{x}^2 = 4(\bar{y} - 2a)^3$$

Replace $\bar{x}$ by x and $\bar{y}$ by y we get $27ax^2 = 4(y - 2a)^3$, which is the evolute of the parabola $x^2 = 4ay$

Example 3: Show that the evolute of the cycloid $x = a(\theta - \sin\theta)$**,** $y = a(1 - \cos\theta)$ **is another cycloid** $x = a(\theta + \sin\theta)$**,** $y = -a(1 - \cos\theta)$ **(Jan-2011)**

Solution:

The equation of the cycloid is given by $x = a(\theta - \sin\theta)$, $y = a(1 - \cos\theta)$

$$\frac{dx}{d\theta} = a(1 - \cos\theta) \qquad \frac{dy}{d\theta} = a\sin\theta$$

$$\frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} = \frac{a\sin\theta}{a(1 - \cos\theta)}$$

$$= \frac{2a\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)}{2a\sin^2\left(\frac{\theta}{2}\right)} = \frac{\cos\left(\frac{\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}$$

$$y_1 = \cot\left(\frac{\theta}{2}\right)$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{d\theta}\left(\frac{dy}{dx}\right)\frac{d\theta}{dx}$$

$$= \frac{d}{d\theta}\left(\cot\left(\frac{\theta}{2}\right)\right)\frac{d\theta}{dx} = -\operatorname{cosec}^2\left(\frac{\theta}{2}\right)\left(\frac{1}{2}\right)\frac{1}{a(1 - \cos\theta)}$$

$$= -\operatorname{cosec}^2\left(\frac{\theta}{2}\right) \cdot \frac{1}{2} \cdot \frac{1}{2a\sin^2\left(\frac{\theta}{2}\right)}$$

$$= -\frac{1}{4a}\operatorname{cosec}^2\left(\frac{\theta}{2}\right)\operatorname{cosec}^2\left(\frac{\theta}{2}\right)$$

$$y_2 = -\frac{1}{4a}\operatorname{cosec}^4\left(\frac{\theta}{2}\right)$$

$$\bar{x} = x - \frac{y_1(1+y_1^2)}{y_2}$$

$$= a(\theta - \sin\theta) - \frac{\cot\left(\frac{\theta}{2}\right)\left(1+\cot^2\left(\frac{\theta}{2}\right)\right)}{-\frac{1}{4a}\operatorname{cosec}^4\left(\frac{\theta}{2}\right)}$$

$$= a(\theta - \sin\theta) + \frac{4a\cot\left(\frac{\theta}{2}\right)\operatorname{cosec}^2\left(\frac{\theta}{2}\right)}{\operatorname{cosec}^4\left(\frac{\theta}{2}\right)}$$

$$= a(\theta - \sin\theta) + \frac{4a\cot\left(\frac{\theta}{2}\right)}{\operatorname{cosec}^2\left(\frac{\theta}{2}\right)} = a(\theta - \sin\theta) + \frac{4a\dfrac{\cos\left(\frac{\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}}{\dfrac{1}{\sin^2\left(\frac{\theta}{2}\right)}}$$

$$= a(\theta - \sin\theta) + 4a\frac{\cos\left(\frac{\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)} x \sin^2\left(\frac{\theta}{2}\right) = a(\theta - \sin\theta) + 4a\cos\left(\frac{\theta}{2}\right)\sin\left(\frac{\theta}{2}\right)$$

$$= a(\theta - \sin\theta) + 2a\sin\theta \qquad \left(\because 2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right) = \sin\theta\right)$$

$$= a\theta - a\sin\theta + 2a\sin\theta \ = a\theta + a\sin\theta$$

$$\bar{x} = a(\theta + \sin\theta)$$

$$\bar{y} = y + \frac{(1+y_1^2)}{y_2}$$

$$= a(1-\cos\theta) + \frac{\left(1+\cot^2\left(\frac{\theta}{2}\right)\right)}{\left(-\frac{1}{4a}\cos ec^4\left(\frac{\theta}{2}\right)\right)} = a(1-\cos\theta) - 4a\frac{\cos ec^2\left(\frac{\theta}{2}\right)}{\cos ec^4\left(\frac{\theta}{2}\right)}$$

$$= a(1-\cos\theta) - 4a\frac{1}{\cos ec^2\left(\frac{\theta}{2}\right)} = a(1-\cos\theta) - 4a\sin^2\left(\frac{\theta}{2}\right)$$

$$= a(1-\cos\theta) - 2a(1-\cos\theta) \qquad \left[\begin{array}{l}\because \sin^2\theta = 1-\cos^2\theta \\ 2\sin^2\left(\frac{\theta}{2}\right) = 1-\cos\theta\end{array}\right]$$

$$= -a(1-\cos\theta)$$

$$\bar{y} = a(\cos\theta - 1)$$

The locus of $(\bar{x}, \bar{y})$ is obtained by replacing $\bar{x}$ by x and $\bar{y}$ by y, we get $x = a(\theta + \sin\theta)$, $y = -a(1-\cos\theta)$ which are the parametric equations of another cycloid.

Example 4: Find the equation of the evolute of the curve

$x = a(\cos\theta + \theta\sin\theta), y = a(\sin\theta - \theta\cos\theta)$ **(Feb-2000, 2006, 2007, May-2002, 2007)**

Solution :

Given that, $x = a(\cos\theta + \theta\sin\theta)$

$$\frac{dx}{d\theta} = a(-\sin\theta + \sin\theta + \theta\cos\theta)$$

$$= a\theta\cos\theta$$

$$y = a(\sin\theta - \theta\cos\theta)$$

$$\frac{dy}{d\theta} = a(\cos\theta + \theta\sin\theta - \cos\theta)$$

$$= a\theta\sin\theta$$

$$\frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} = \frac{a\theta\sin\theta}{a\theta\cos\theta}$$

$$y_1 = \tan\theta$$

$$\frac{d^2y}{dx^2} = \frac{d}{d\theta}\left(\frac{dy}{dx}\right)\frac{d\theta}{dx} = \frac{d}{d\theta}(\tan\theta)\frac{d\theta}{dx}$$

$$= \sec^2\theta\frac{1}{a\theta\cos\theta}$$

$$y_2 = \frac{1}{a\theta\cos^3\theta}$$

$$\bar{x} = x - \frac{y_1(1+y_1^2)}{y_2}$$

$$= a(\cos\theta + \theta\sin\theta) - \frac{\tan\theta\left(1+\tan^2\theta\right)}{\left(\frac{1}{a\theta\cos^3\theta}\right)}$$

$$= a(\cos\theta + \theta\sin\theta) - a\theta\tan\theta\sec^2\theta\cos^3\theta$$

$$= a\cos\theta + a\theta\sin\theta - a\theta\frac{\sin\theta}{\cos\theta}\frac{1}{\cos^2\theta}\cos^3\theta$$

$$= a\cos\theta + a\theta\sin\theta - a\theta\sin\theta$$

$$\overline{x} = a\cos\theta \qquad ...(1)$$

$$\overline{y} = y + \left(\frac{1+y_1^2}{y_2}\right)$$

$$= a(\sin\theta - \theta\cos\theta) + \frac{(1+\tan^2\theta)}{\left(\frac{1}{a\theta\cos^3\theta}\right)}$$

$$= a\sin\theta - a\theta\cos\theta + \sec^2\theta a\theta\cos^3\theta$$

$$= a\sin\theta - a\theta\cos\theta + a\theta\cos\theta$$

$$\overline{y} = a\sin\theta \qquad ...(2)$$

From (1) & (2) $\overline{x}^2 + \overline{y}^2 = a^2\cos^2\theta + a^2\sin^2\theta$

$$= a^2(\cos^2\theta + \sin^2\theta)$$

$$\overline{x}^2 + \overline{y}^2 = a^2$$

Replacing $\overline{x}, \overline{y}$ by x, y respectively, we get the evolute of the given curve as $x^2 + y^2 = a^2$

Example 5: Show that the evolute of the tractrix $x = a\left(\cos t + \log\tan\frac{t}{2}\right)$, $y = a\sin t$ **at** t **is a curve (catenary)** $y = a\cosh\left(\frac{x}{a}\right)$ **(Nov-2008)**

Solution:

Given that $x = a\left(\cos t + \log\tan\frac{t}{2}\right)$

$$\frac{dx}{dt} = a\left[-\sin t + \frac{1}{\tan\left(\frac{t}{2}\right)}\sec^2\left(\frac{t}{2}\right)\frac{1}{2}\right]$$

$$= a\left[-\sin t + \frac{\cos\left(\frac{t}{2}\right)}{\sin\left(\frac{t}{2}\right)}\frac{1}{\cos^2\left(\frac{t}{2}\right)}\frac{1}{2}\right] = a\left[-\sin t + \frac{1}{2\sin\left(\frac{t}{2}\right)\cos\left(\frac{t}{2}\right)}\right]$$

$$= a\left[-\sin t + \frac{1}{\sin t}\right] = a\left[\frac{-\sin^2 t + 1}{\sin t}\right]$$

$$\frac{dx}{dt} = a\frac{\cos^2 t}{\sin t}$$

$$y = a\sin t$$

$$\frac{dy}{dt} = a\cos t$$

$$\therefore \frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)} = \frac{a\cos t}{\left(\frac{a\cos^2 t}{\sin t}\right)} = \cos t\frac{\sin t}{\cos^2 t}$$

$$\frac{dy}{dx} = \frac{\sin t}{\cos t}$$

$$y_1 = \tan t$$

$$\frac{d^2 y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}(\tan t)$$

$$= \frac{d}{dt}(\tan t)\frac{dt}{dx} = \sec^2 t\frac{\sin t}{a\cos^2 t}$$

$$y_2 = \frac{\sin t}{a\cos^4 t}$$

$$\bar{x} = x - \left[\frac{y_1(1+y_1^2)}{y_2}\right]$$

$$= a\left(\cos t + \log\tan\left(\frac{t}{2}\right)\right) - \left[\frac{\tan t(1+\tan^2 t)}{\left(\dfrac{\sin t}{a\cos^4 t}\right)}\right]$$

$$= a\cos t + a\log\tan\left(\frac{t}{2}\right) - \frac{\sin t}{\cos t}\,\frac{\sec^2 t}{\sin t}\,a\cos^4 t$$

$$= a\cos t + a\log\tan\left(\frac{t}{2}\right) - a\cos t$$

$$\overline{x} = a\log\tan\left(\frac{t}{2}\right) \qquad \text{...(1)}$$

$$\overline{y} = y + \left[\frac{1+y_1^2}{y_2}\right]$$

$$= a\sin t + \left[\frac{1+\tan^2 t}{\left(\dfrac{\sin t}{a\cos^4 t}\right)}\right] = a\sin t + \sec^2 t\,\frac{a\cos^4 t}{\sin t}$$

$$= a\sin t + \frac{a\cos^2 t}{\sin t} = \frac{a\sin^2 t + a\cos^2 t}{\sin t}$$

$$\overline{y} = \frac{a}{\sin t} \qquad \text{...(2)}$$

From(1), $\overline{x} = a\log\tan\left(\frac{t}{2}\right)$

$$\frac{\overline{x}}{a} = \log\tan\left(\frac{t}{2}\right)$$

$$e^{\frac{\overline{x}}{a}} = \tan\left(\frac{t}{2}\right)$$

$$e^{-\frac{\bar{x}}{a}} = \frac{1}{\tan\left(\frac{t}{2}\right)}$$

We know that, $\dfrac{e^{x}+e^{-x}}{2} = \cosh x$

$$\therefore \frac{e^{-\frac{\bar{x}}{a}}+e^{-\frac{\bar{x}}{a}}}{2} = \cosh\frac{\bar{x}}{a}$$

$$\text{(i.e.) } \cosh\frac{\bar{x}}{a} = \frac{\tan\left(\frac{t}{2}\right)+\frac{1}{\tan\left(\frac{t}{2}\right)}}{2}$$

$$= \frac{\tan^{2}\left(\frac{t}{2}\right)+1}{2\tan\left(\frac{t}{2}\right)} = \frac{\sec^{2}\left(\frac{t}{2}\right)}{2\tan\left(\frac{t}{2}\right)}$$

$$= \frac{1}{2\cos^{2}\left(\frac{t}{2}\right)}\frac{\cos\left(\frac{t}{2}\right)}{\sin\left(\frac{t}{2}\right)} = \frac{1}{2\sin\left(\frac{t}{2}\right)\cos\left(\frac{t}{2}\right)}$$

$$\cosh\left(\frac{\bar{x}}{a}\right) = \frac{1}{\sin t} \qquad \text{(by equation (2))} \quad [\therefore \bar{y} = \frac{a}{\sin t} \qquad \frac{\bar{y}}{a} = \frac{1}{\sin t}]$$

$$\cosh\left(\frac{\bar{x}}{a}\right) = \frac{\bar{y}}{a}$$

$$\bar{y} = a\cosh\left(\frac{\bar{x}}{a}\right)$$

Replace $\bar{x}$ by x and $\bar{y}$ by y we get, $y = a\cosh\left(\frac{x}{a}\right)$ as the evolute of the given tractrix.

Hence proved.

Example 6: Show that the evolute of the rectangular hyperbola $xy = c^2$ is

$(x+y)^{\frac{2}{3}} - (x-y)^{\frac{2}{3}} = (4c)^{\frac{2}{3}}$

Solution:

Given that $xy = c^2$

The parametric coordinates of any point on the rectangular hyperbola are $x = ct$, $y = \dfrac{c}{t}$

$$\frac{dx}{dt} = c \text{ and } \frac{dy}{dt} = -\frac{c}{t^2}$$

$$\frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)} = \frac{\left(\frac{-c}{t^2}\right)}{c}$$

$$y_1 = \frac{-1}{t^2}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dt}\left(\frac{dy}{dx}\right)\frac{dt}{dx}$$

$$= \frac{d}{dt}\left(\frac{-1}{t^2}\right)\frac{1}{c} = \frac{2}{t^3}\frac{1}{c}$$

$$y_2 = \frac{2}{ct^3}$$

$$\bar{x} = x - y_1\left[\frac{\left(1+y_1^2\right)}{y_2}\right]$$

$$= ct - \left(-\frac{1}{t^2}\right)\left[\frac{1+\frac{1}{t^4}}{\left(\frac{2}{ct^3}\right)}\right] = ct + \frac{1}{t^2}\frac{\left(t^4+1\right)ct^3}{2t^4}$$

$$= ct + \frac{(1+t^4)c}{2t^3}$$

$$\overline{x} = \frac{2ct^4 + c + t^4 c}{2t^3} \quad = \frac{3ct^4 + c}{2t^3}$$

$$= \frac{c\left(3t^4 + 1\right)}{2t^3} \qquad \ldots(1)$$

$$\overline{y} = y + \left[\frac{1 + {y_1}^2}{y_2}\right]$$

$$= \frac{c}{t} + \left[\frac{1 + \frac{1}{t^4}}{\frac{2}{ct^3}}\right] = \frac{c}{t} + \frac{ct^3(t^4 + 1)}{2t^4}$$

$$= \frac{2c + c + ct^4}{2t} = \frac{3c + ct^4}{2t}$$

$$\overline{y} = \frac{c(3 + t^4)}{2t} \qquad \ldots(2)$$

Now $\overline{x} + \overline{y} = \dfrac{c(3t^4 + 1)}{2t^3} + \dfrac{c(3 + t^4)}{2t}$

$$= \frac{3ct^4 + c + 3ct^2 + ct^6}{2t^3} = \frac{c}{2t^3}\left[3t^4 + 1 + 3t^2 + t^6\right]$$

$$\overline{x} + \overline{y} = \frac{c}{2t^3}\left[1 + t^2\right]^3$$

$$(\overline{x} + \overline{y})^{\frac{2}{3}} = \left(\frac{c}{2t^3}\right)^{\frac{2}{3}} \left(\left(1 + t^2\right)^3\right)^{\frac{2}{3}}$$

$$= \left(\frac{c}{2}\right)^{\frac{2}{3}} \frac{1}{t^2}\left(1 + t^2\right)^2 \qquad \ldots(3)$$

$$\bar{x}-\bar{y}=\frac{c(3t^4+1)}{2t^3}-\frac{c(3+t^4)}{2t}$$

$$=\frac{3ct^4+c-3ct^2-ct^6}{2t^3}\quad=\frac{c}{2t^3}(3t^4+1-3t^2-t^6)$$

$$=\frac{c}{2t^3}(1-t^2)^3$$

$$(\bar{x}-\bar{y})^{\frac{2}{3}}=\left(\frac{c}{2}\right)^{\frac{2}{3}}\frac{1}{t^2}(1-t^2)^2 \qquad \ldots(4)$$

(3) and (4) becomes,

$$(\bar{x}+\bar{y})^{\frac{2}{3}}-(\bar{x}-\bar{y})^{\frac{2}{3}}=\left(\frac{c}{2}\right)^{\frac{2}{3}}\frac{1}{t^2}\left[\left(1+t^2\right)^2-\left(1-t^2\right)^2\right]$$

$$=\left(\frac{c}{2}\right)^{\frac{2}{3}}\frac{1}{t^2}4t^2=4\left(\frac{c}{2}\right)^{\frac{2}{3}}$$

$\therefore$ Replacing $\bar{x}$ by x and $\bar{y}$ by y , we get $(\bar{x}+\bar{y})^{\frac{2}{3}}-(\bar{x}-\bar{y})^{\frac{2}{3}}=(4c)^{\frac{2}{3}}$ where $c=\frac{c}{2}$

This is the required equation of evolute of the rectangular hyperbola.

Example 7: Find the evolute of the curve $x^{\frac{2}{3}}+y^{\frac{2}{3}}=a^{\frac{2}{3}}$

(April-2003,2007, Nov-2003)

Solution:

The Parametric coordinates of any point on the curve are $x=a\cos^3\theta$, $y=a\sin^3\theta$

$$\frac{dx}{d\theta}=3a\cos^2\theta(-\sin\theta) \qquad \frac{dy}{d\theta}=3a\sin^2\theta(\cos\theta)$$

$$=-3a\sin\theta\cos^2\theta$$

$$\therefore \frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} = \frac{3a\sin^2\theta\cos\theta}{-3a\sin\theta\cos^2\theta} = \frac{-\sin\theta}{\cos\theta}$$

$$= -\tan\theta$$

$$\frac{d^2y}{dx^2} = \frac{d}{d\theta}\left(\frac{dy}{dx}\right)\left(\frac{d\theta}{dx}\right)$$

$$= \frac{d}{d\theta}(-\tan\theta) \times \left(\frac{1}{-3a\cos^2\theta\sin\theta}\right)$$

$$= -\sec^2\theta \times \left(\frac{1}{-3a\cos^2\theta\sin\theta}\right)$$

$$y_2 = \frac{1}{3a\sin\theta\cos^4\theta}$$

$$\overline{x} = x - y_1\left[\frac{1+y_1^2}{y_2}\right]$$

$$= a\cos^3\theta - \left[\frac{-\tan\theta\left(1+\tan^2\theta\right)}{\frac{1}{3a\sin\theta\cos^4\theta}}\right]$$

$$= a\cos^3\theta + \tan\theta\sec^2\theta \times 3a\sin\theta\cos^4\theta$$

$$= a\cos^3\theta + \frac{\sin\theta}{\cos\theta} \times \frac{1}{\cos^2\theta} \times 3a\sin\theta\cos^4\theta$$

$$= a\cos^3\theta + 3a\sin^2\theta\cos\theta$$

$$\overline{x} = a\cos^3\theta + 3a\sin^2\theta\cos\theta \qquad \ldots(1)$$

$$\overline{y} = y + \left[\frac{1+y_1^2}{y_2}\right]$$

$$= a\sin^3\theta + \left[\frac{1+\tan^2\theta}{\frac{1}{3a\sin\theta\cos^4\theta}}\right]$$

$$= a\sin^3\theta + \sec^2\theta \times 3a\sin\theta\cos^4\theta$$

$$\bar{y} = a\sin^3\theta + 3a\sin\theta\cos^2\theta \qquad \ldots(2)$$

(1) + (2)gives,

$$\bar{x} + \bar{y} = a\cos^3\theta + 3a\sin^2\theta\cos\theta + a\sin^3\theta + 3a\sin\theta\cos^2\theta$$

$$= a\left[\cos^3\theta + 3\sin^2\theta\cos\theta + 3\sin\theta\cos^2\theta + \sin^3\theta\right]$$

$$\bar{x} + \bar{y} = a\left(\cos\theta + \sin\theta\right)^3 \qquad \ldots(3)$$

(1) - (2) gives,

$$\bar{x} - \bar{y} = a\cos^3\theta - a\sin^3\theta + 3a\sin^2\theta\cos\theta - 3a\sin\theta\cos^2\theta$$

$$= a(\cos\theta - \sin\theta)^3 \qquad \ldots(4)$$

Now $(\bar{x}+\bar{y})^{\frac{2}{3}} + (\bar{x}-\bar{y})^{\frac{2}{3}} = \left(a\right)^{\frac{2}{3}}\left(\cos\theta+\sin\theta\right)^2 + \left(a\right)^{\frac{2}{3}}\left(\cos\theta-\sin\theta\right)^2$

$$= \left(a\right)^{\frac{2}{3}}\left[\cos^2\theta + \sin^2\theta + 2\sin\theta\cos\theta + \cos^2\theta + \sin^2\theta - 2\sin\theta\cos\theta\right]$$

$$= \left(a\right)^{\frac{2}{3}}(1+1)$$

$$(\bar{x}+\bar{y})^{\frac{2}{3}} + (\bar{x}-\bar{y})^{\frac{2}{3}} = 2a^{2/3}$$

Replacing $\bar{x}$ by x and $\bar{y}$ by y, we get $(x+y)^{\frac{2}{3}} + (x-y)^{\frac{2}{3}} = 2a^{\frac{2}{3}}$

This is the required equation of the evolute of the given curve.

Example 8: Find the evolute of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(Nov-2002,2005,April-2003,2004)

Solution:

The parametric coordinates of any point on the ellipse are $x = a\cos\theta, y = b\sin\theta$

$$\frac{dx}{d\theta} = -a\sin\theta \qquad \frac{dy}{d\theta} = b\cos\theta$$

$$\therefore \frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} = \frac{b\cos\theta}{-a\sin\theta}$$

$$y_1 = -\frac{b}{a}\cot\theta$$

$$\frac{d^2y}{dx^2} = \frac{d}{d\theta}\left(\frac{dy}{dx}\right)\frac{d\theta}{dx}$$

$$= \frac{d}{d\theta}\left(\frac{-b}{a}\cot\theta\right) \times \frac{1}{-a\sin\theta}$$

$$= \frac{b}{a}\cos ec^2\theta\left(\frac{-1}{a\sin\theta}\right)$$

$$y_2 = -\frac{b}{a^2}\cos ec^3\theta$$

$$\bar{x} = x - y_1\left[\frac{\left(1+y_1^2\right)}{y_2}\right]$$

$$= a\cos\theta - \frac{\left(\frac{-b}{a}\cot\theta\right)\left(1+\frac{b^2}{a^2}\cot^2\theta\right)}{\left(\frac{-b}{a^2}\cos ec^3\theta\right)}$$

$$= a\cos\theta - \left(\frac{b}{a}\cot\theta\right) \times \left(\frac{a^2}{b\cos ec^3\theta}\right) \times \left(\frac{a^2 + b^2\cot^2\theta}{a^2}\right)$$

$$= a\cos\theta - \frac{a^2 b}{a^3 b}\frac{\cos\theta}{\sin\theta}\sin^3\theta\left(a^2 + b^2\frac{\cos^2\theta}{\sin^2\theta}\right)$$

$$= a\cos\theta - \frac{1}{a}\cos\theta\sin^2\theta\left(a^2 + b^2\frac{\cos^2\theta}{\sin^2\theta}\right)$$

$$= a\cos\theta - \frac{\cos\theta\sin^2\theta}{a}\left(\frac{a^2\sin^2\theta + b^2\cos^2\theta}{\sin^2\theta}\right)$$

$$= a\cos\theta - \frac{a^2\cos\theta\sin^2\theta}{a} - \frac{b^2\cos^3\theta}{a}$$

$$= a\cos\theta(1-\sin^2\theta) - \frac{b^2}{a}\cos^3\theta$$

$$= a\cos\theta\cos^2\theta - \frac{b^2}{a}\cos^3\theta$$

$$= a\cos^3\theta - \frac{b^2}{a}\cos^3\theta$$

$$= \frac{a^2\cos^3\theta - b^2\cos^3\theta}{a}$$

$$\bar{x} = \frac{\cos^3\theta}{a}\left(a^2 - b^2\right) \qquad \ldots(1)$$

$$\bar{y} = y + \left(\frac{1+y_1^2}{y_2}\right)$$

$$= b\sin\theta + \left[\frac{1+\frac{b^2}{a^2}\cot^2\theta}{-\frac{b}{a^2}\cos ec^3\theta}\right]$$

$$= b\sin\theta - \frac{\left(a^2 + b^2\cot^2\theta\right)}{a^2}\times\frac{a^2}{b\cos ec^3\theta}$$

$$= b\sin\theta - \left(\frac{a^2 + b^2 \dfrac{\cos^2\theta}{\sin^2\theta}}{b \dfrac{1}{\sin^3\theta}} \right)$$

$$= b\sin\theta - \left(\frac{a^2 \sin^2\theta + b^2 \cos^2\theta}{b\sin^2\theta} \right) \sin^3\theta$$

$$= b\sin\theta - \frac{a^2 \sin^3\theta - b^2\cos^2\theta\sin\theta}{b}$$

$$= \frac{b^2\sin\theta - a^2\sin^3\theta - b^2\cos^2\theta\sin\theta}{b}$$

$$= \frac{b^2\sin\theta(1-\cos^2\theta) - a^2\sin^3\theta}{b}$$

$$= \frac{b^2\sin^3\theta - a^2\sin^3\theta}{b}$$

$$\bar{y} = \frac{\sin^3\theta}{b}(b^2 - a^2) \qquad \ldots(2)$$

From (1), $a\bar{x} = (a^2 - b^2)\cos^3\theta$

$$\frac{a\bar{x}}{a^2 - b^2} = \cos^3\theta \qquad \ldots(3)$$

From (2), $b\bar{y} = (b^2 - a^2)\sin^3\theta$

$$\frac{b\bar{y}}{b^2 - a^2} = \sin^3\theta \qquad \ldots(4)$$

$$(3) \Rightarrow \left(\cos^3\theta\right)^{\frac{2}{3}} = \left(\frac{a\bar{x}}{a^2 - b^2} \right)^{\frac{2}{3}}$$

$$(4) \Rightarrow \left(\sin^3\theta\right)^{\frac{2}{3}} = \left(\frac{b\bar{y}}{b^2 - a^2} \right)^{\frac{2}{3}}$$

(i.e.) $\cos^2\theta = \left(\dfrac{a\overline{x}}{a^2-b^2}\right)^{\frac{2}{3}}$, $\sin^2\theta = \left(\dfrac{b\overline{y}}{b^2-a^2}\right)^{\frac{2}{3}}$

We know that, $\sin^2\theta + \cos^2\theta = 1$

$$\left(\frac{b\overline{y}}{b^2-a^2}\right)^{\frac{2}{3}} + \left(\frac{a\overline{x}}{a^2-b^2}\right)^{\frac{2}{3}} = 1$$

$$\left(b\overline{y}\right)^{\frac{2}{3}} + \left(a\overline{x}\right)^{\frac{2}{3}} = (a^2-b^2)^{\frac{2}{3}}$$

Replacing $\overline{x}\ by\ x$ & $\overline{y}\ by\ y$, we get $\left(ax\right)^{\frac{2}{3}} + \left(by\right)^{\frac{2}{3}} = (a^2-b^2)^{\frac{2}{3}}$

Which is the equation of the evolute of the ellipse.

Example 9: Find the evolute of the hyperbola $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$ **(Jan - 2000, 2003)**

Solution:

The parametric equation of any point on the hyperbola are

$$x = a\sec\theta \quad and \quad y = b\tan\theta$$

$$\frac{dx}{d\theta} = a\sec\theta\tan\theta, \ \frac{dy}{d\theta} = b\sec^2\theta$$

$$\therefore \frac{dy}{dx} = \frac{\left(\dfrac{dy}{d\theta}\right)}{\left(\dfrac{dx}{d\theta}\right)} = \frac{b\sec^2\theta}{a\sec\theta\tan\theta}$$

$$= \frac{b}{a}\frac{1}{\cos\theta}\frac{\cos\theta}{\sin\theta} = \frac{b}{a\sin\theta}$$

$$y_1 = \frac{b}{a}\cos ec\theta$$

$$\frac{d^2y}{dx^2} = \frac{d}{d\theta}\left(\frac{dy}{dx}\right)\frac{d\theta}{dx} = \frac{d}{d\theta}\left(\frac{b}{a}\cos ec\theta\right)\frac{d\theta}{dx}$$

$$=\frac{b}{a}\left(-\cos ec\theta\cot\theta\right)\times\frac{1}{a\sec\theta\tan\theta}$$

$$=\frac{-b}{a}\frac{1}{\sin\theta}\frac{\cos\theta}{\sin\theta}\times\frac{1}{a}\times\cos\theta\times\frac{\cos\theta}{\sin\theta}$$

$$=\frac{-b}{a^2}\frac{\cos^3\theta}{\sin^3\theta}$$

$$y_2=-\frac{b}{a^2}\cot^3\theta$$

$$\overline{x}=x-\frac{y_1(1+y_1^2)}{y_2}$$

$$=a\sec\theta-\frac{\frac{b}{a}\cos ec\theta\left[1+\frac{b^2}{a^2}\cos ec^2\theta\right]}{\frac{-b}{a^2}\cot^3\theta}$$

$$=asec\theta+\frac{b}{a}\cos ec\theta\frac{\left(a^2+b^2\cos ec^2\theta\right)}{a^2}\times\frac{a^2}{b}\frac{\sin^3\theta}{\cos^3\theta}$$

$$=\frac{a}{\cos\theta}+\frac{1}{a}\frac{1}{\sin\theta}\left(a^2+b^2\frac{1}{\sin^2\theta}\right)\frac{\sin^3\theta}{\cos^3\theta}$$

$$=\frac{a}{\cos\theta}+\frac{1}{a\sin\theta}\left(\frac{a^2\sin^2\theta+b^2}{\sin^2\theta}\right)\frac{\sin^3\theta}{\cos^3\theta}$$

$$=\frac{a}{\cos\theta}+\frac{a^2\sin^2\theta+b^2}{a\cos^3\theta}$$

$$=\frac{a^2\cos^2\theta+a^2\sin^2\theta+b^2}{a\cos^3\theta}$$

$$=\frac{a^2(\cos^2\theta+\sin^2\theta)+b^2}{a\cos^3\theta}$$

$$\bar{x} = \frac{a^2 + b^2}{a\cos^3\theta} \qquad \ldots(1)$$

$$\bar{y} = y + \left(\frac{1 + y_1^2}{y_2}\right) = b\tan\theta + \left[\frac{1 + \frac{b^2}{a^2}\cos ec^2\theta}{-\frac{b}{a^2}\cot^3\theta}\right]$$

$$= b\tan\theta + \left(-\frac{1}{b}\right)\frac{\sin^3\theta}{\cos^3\theta}\left(a^2 + b^2\frac{1}{\sin^2\theta}\right)$$

$$= \frac{b\sin\theta}{\cos\theta} + \left(-\frac{1}{b}\right)\left(\frac{a^2\sin^3\theta}{\cos^3\theta} + \frac{b^2\sin\theta}{\cos^3\theta}\right)$$

$$= \frac{b\sin\theta}{\cos\theta} - \left(a^2\sin^2\theta + b^2\right)\frac{\sin\theta}{b\cos^3\theta}$$

$$= \frac{b^2\sin\theta\cos^2\theta - a^2\sin^3\theta - b^2\sin\theta}{b\cos^3\theta}$$

$$= \frac{b^2\sin\theta(1 - \sin^2\theta) - a^2\sin^3\theta - b^2\sin\theta}{b\cos^3\theta}$$

$$= \frac{1}{b\cos^3\theta}\left[b^2\sin\theta - b^2\sin^3\theta - a^2\sin^3\theta - b^2\sin\theta\right]$$

$$= -\frac{1}{b\cos^3\theta}\left[\left(a^2 + b^2\right)\sin^3\theta\right]$$

$$\bar{y} = -\frac{\left(a^2 + b^2\right)\tan^3\theta}{b} \qquad \ldots(2)$$

From (1), $\quad \bar{x} = \sec^3\theta\left(\frac{a^2 + b^2}{a}\right)$

$$\sec^3\theta = \left(\frac{a\bar{x}}{a^2 + b^2}\right), \quad \sec^2\theta = \left(\frac{a\bar{x}}{a^2 + b^2}\right)^{\frac{2}{3}}$$

From (2), $\overline{y}=\left(\dfrac{-\left(a^2+b^2\right)}{b}\right)\tan^3\theta$

$$\tan^3\theta=-\frac{b\overline{y}}{a^2+b^2} \quad , \quad \tan^2\theta=\left(-\frac{b\overline{y}}{a^2+b^2}\right)^{\frac{2}{3}}$$

We know that, $\sec^2\theta-\tan^2\theta=1$

$$\left(\frac{a\overline{x}}{a^2+b^2}\right)^{\frac{2}{3}}-\left(\frac{b\overline{y}}{a^2+b^2}\right)^{\frac{2}{3}}=1$$

$$\left(\frac{a\overline{x}}{a^2+b^2}\right)^{\frac{2}{3}}-\left(\frac{b\overline{y}}{a^2+b^2}\right)^{\frac{2}{3}}=1$$

Replacing $\overline{x}$ *by* x & $\overline{y}$ *by* y, we get $\left(a\overline{x}\right)^{\frac{2}{3}}-\left(b\overline{y}\right)^{\frac{2}{3}}=(a^2+b^2)^{\frac{2}{3}}$ Which is the required equation of the evolute of the hyperbola.

Exercise

1. Prove that the evolute of the curve x=ct , y=$\dfrac{c}{t}$ is $\left(x+y\right)^{\frac{2}{3}}+\left(x-y\right)^{\frac{2}{3}}=(4c)^{\frac{2}{3}}$

2. Find the equation of parametric form of the evolute of the curve $x=2\cos t+\cos 2t$, $y=2\sin t+\sin 2t$

3. Show that the evolute of the cycloid $x=a(1-\sin\theta)$, $y=a(1-\cos\theta)$ is another cycloid given by $x=a(\theta-\sin\theta)$, $y-2a=a(1+\cos\theta)$

Answers

2. $\overline{x}=\dfrac{2}{3}\cos t-\dfrac{1}{3}\cos 2t,\ \overline{y}=\dfrac{2}{3}\sin t-\dfrac{\sin 2t}{3}$

1.8. Gamma and Beta function

Improper Integrals

The integral $\int_a^b f(x)dx$ is called an improper integral if any one of the following conditions is satisfied.

(i) $a = -\infty$ or $b = \infty$ or both (i.e.) one or both the limits of integration is infinite.

(ii) $f(x)$ is unbounded at one or more points in the interval $a \le x \le b$

Integrals corresponding to (i) and (ii) are called improper integrals of the first and second kind respectively. Integrals with both conditions (i) and (ii) are called improper integrals of the third kind.

Examples: 1. $\int_0^\infty \frac{dx}{1+x^2}$ is an improper integral of the first kind.

2. $\int_0^1 \frac{dx}{1-x^2}$ is an improper integral of the second kind

3. $\int_0^\infty e^{-x}x^{n-1}dx$ Where $n > o$ is an improper integral of the third kind.

1.8.1. Gamma Function

The Gamma Function (or) the Generalized Factorial Function denoted by $\Gamma(n)$ is defined as the definite integral.

$$\Gamma(n) = \int_0^\infty e^{-x}x^{n-1}dx \text{ where } n > o$$

$$\Gamma(n+1) = \int_0^\infty e^{-x}x^{n}dx, \quad n+1 > 0 \text{ or } n > -1$$

1.8.2. Properties of Gamma function.

Property 1: $\Gamma(1)=1$

We know that $\Gamma(n)=\int_0^\infty e^{-x}x^{n-1}dx,$

Put $n=1$

$$\Gamma(1)=\int_0^\infty e^{-x}x^{1-1}dx \; =\int_0^\infty e^{-x} \; =\left[-e^{-x}\right]_0^\infty$$

$$\Gamma(1)=1$$

Property 2: Recurrence Formula (or) Reduction Formula for $\Gamma(n)$ **(May- 2002)**

The reduction or recurrence formula is given by $\Gamma(n+1)=n\Gamma(n)$

We know that $\Gamma(n)=\int_0^\infty e^{-x}x^{n-1}dx, \quad n>o$

Replace n by $n+1$, we get

$$\Gamma(n+1)=\int_0^\infty e^{-x}x^{n+1-1}dx,$$

$$\int_0^\infty e^{-x}x^n dx=\int_0^\infty x^n d(-e^{-x})=(-x^n e^{-x})_0^\infty-\int_0^\infty -e^{-x}d(x^n)$$

$$=\int_0^\infty e^{-x}nx^{n-1}dx=n\int_0^\infty e^{-x}x^{n-1}dx$$

$$\Gamma(n+1)=n\Gamma n$$

$$\therefore \Gamma(n)=\frac{\Gamma(n+1)}{n}$$

Corollary

If n is a positive integer, then $\Gamma(n+1)=n!$

We know that $\Gamma(n+1)=n\Gamma n$ …(1)

Put $n = n-1$ in (1)

$$\Gamma(n) = (n-1)\Gamma(n-1) \qquad \text{...(2)}$$

Put $n = n-2$ in (1)

$$\Gamma(n-1) = (n-2)\Gamma(n-2) \qquad \text{…(3)}$$

Put $n = n-3$ in (1)

$$\Gamma(n-2) = (n-3)\Gamma(n-3) \qquad \text{…(4)}$$

…

…

Substituting (2), (3), (4) in (1), we get

$$\Gamma(n+1) = n\Gamma(n)$$

$$= n(n-1)\Gamma(n-1)$$

$$= n(n-1)(n-2)\Gamma(n-2)$$

$$= n(n-1)(n-2)(n-3)\Gamma(n-3)$$

$$\vdots$$

$$= n(n-1)(n-2)(n-3)\Gamma(n-3)\ \ldots\ 1\Gamma(1)$$

$$= n(n-1)(n-2)(n-3)\ \ldots\ 1$$

$= n!$ Where n is a positive integer.

Property 3: $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

We know that $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx$

Put $n = \frac{1}{2}$

$$\Gamma\left(\frac{1}{2}\right) = \int_0^\infty e^{-x} x^{\frac{1}{2}-1} dx$$

Put $x = y^2$ when $x = \infty$, $y = \infty$

$dx = 2ydy$ $\quad$ $x = 0$, $y = 0$

$$\therefore \Gamma\left(\frac{1}{2}\right) = \int_0^\infty e^{-y^2} y^{-1} 2ydy$$

$$= 2\int_0^\infty e^{-y^2} dy \qquad ...(1)$$

$$\Gamma\left(\frac{1}{2}\right) = 2\int_0^\infty e^{-x^2} dx \qquad ...(2)$$

Multiply (1) and (2), we get

$$\left[\Gamma\left(\frac{1}{2}\right)\right]^2 = 4\int_0^\infty e^{-y^2} dy \int_0^\infty e^{-x^2} dx$$

$$= 4\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dxdy \qquad ...(3)$$

The region of integration in the double integration (3) is the infinite area in the first quadrant. We can change to polar coordinates by substituting

$$x = r\cos\theta, \quad y = r\sin\theta, \quad x^2 + y^2 = r^2$$

The element of area $dydx$ becomes $rd\theta dr$

The limits of integration are 0 to ∞ for r and 0 to $\frac{\pi}{2}$ for θ

Hence (3) becomes, $\left[\Gamma\left(\frac{1}{2}\right)\right]^2 = 4\int_0^{\frac{\pi}{2}}\int_0^\infty e^{-r^2} rd\theta dr$

$$\left[\Gamma\left(\frac{1}{2}\right)\right]^2 = 4\int_0^{\frac{\pi}{2}}\left[\int_0^\infty e^{-r^2} rdr\right] d\theta$$

$$= 4\int_0^{\frac{\pi}{2}} \left[\int_0^{\infty} e^{-r^2} d\left(\frac{1}{2}r^2\right)\right] d\theta = 4\times\frac{1}{2}\int_0^{\frac{\pi}{2}} \left[\int_0^{\infty} e^{-r^2} d\left(r^2\right)\right] d\theta$$

$$= 2\int_0^{\frac{\pi}{2}} \left[-e^{-r^2}\right]_0^{\infty} d\theta = 2\int_0^{\frac{\pi}{2}} 1 d\theta = 2\left[\theta\right]_0^{\pi/2} = 2\times\frac{\pi}{2} = \pi$$

$$\left[\Gamma\left(\frac{1}{2}\right)\right]^2 = \pi$$

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

1.8.3. Beta function

The beta function which involves two parameters m and n is defined by the relation.

$$\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}dx, \quad m>0, \quad n>0 \qquad ...(1)$$

The above integral is an improper integral and is convergent when m and n are positive numbers. Beta function is also called the **Eulerian integrals** of the first kind.

1.8.4. Properties of Beta function

Property 1: The function $\beta(m,n)$ is symmetric in its parameters or arguments.

(i.e.) $\beta(m,n) = \beta(n,m)$

We know that

$$\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}dx,$$

Putting $x = 1-y$ when $x = 0, \quad y = 1$

$dx = -dy$ $\quad x = 1, \quad y = 0$

$$\beta(m,n) = \int_1^0 (1-y)^{m-1}(1-1+y)^{n-1}(-dy),$$

$$= -\int_{1}^{0} y^{n-1}(1-y)^{m-1}dy$$

$$= \int_{0}^{1} y^{n-1}(1-y)^{m-1}dy \qquad \left(\int_{a}^{b} f(x)dx = -\int_{b}^{a} f(x)dx\right)$$

$$\beta(m,n) = \int_{0}^{1} x^{n-1}(1-x)^{m-1}dx, \qquad \text{(Replace the dummy variable y by x)}$$

$$\therefore \beta(m,n) = \beta(n,m)$$

Property 2: Beta function in terms of circular functions is given by

$$\beta(m,n) = 2\int_{0}^{\frac{\pi}{2}} \sin^{2m-1}\theta\cos^{2n-1} d\theta$$

We know that

$$\beta(m,n) = \int_{0}^{1} x^{m-1}(1-x)^{n-1}dx$$

…(1)

Putting $x = \sin^2\theta$ when $x = 0,\ \theta = 0$

$$dx = 2\sin\theta\cos\theta d\theta \qquad x = 1, \quad \theta = \frac{\pi}{2}$$

$$\beta(m,n) = \int_{0}^{\frac{\pi}{2}} (\sin^2\theta)^{m-1}(1-\sin^2\theta)^{n-1} 2\sin\theta\cos\theta d\theta$$

$$= 2\int_{0}^{\frac{\pi}{2}} \sin^{2m-2}\theta\cos^{2n-2}\theta\sin\theta\cos\theta d\theta = 2\int_{0}^{\frac{\pi}{2}} \sin^{2m-1}\theta\cos^{2n-1}\theta d\theta$$

$$\beta(m,n) = 2\int_{a}^{\frac{\pi}{2}} \sin^{2m-1}\theta\cos^{2n-1} d\theta$$

Property 3: Another form of Beta function is

$$\beta(m,n)=\int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\,dx, \qquad m>0, \quad n>0$$

We know that $\beta(m,n)=\int_0^1 x^{m-1}(1-x)^{n-1}\,dx$ ... (1)

Putting $x=\frac{1}{1+y}$ when $x=0, \quad y=\infty$

$$dx=\frac{-1}{(1+y)^2}\,dy \qquad x=1, \quad y=0$$

Equation (1) becomes,

$$\beta(m,n)=\int_\infty^0 \left(\frac{1}{1+y}\right)^{m-1}\left(1-\frac{1}{1+y}\right)^{n-1}\left(\frac{-dy}{(1+y)^2}\right)$$

$$=\int_0^\infty \frac{1}{(1+y)^{m-1}}\left(\frac{1+y-1}{1+y}\right)^{n-1}\frac{dy}{(1+y)^2}$$

$$=\int_0^\infty \frac{y^{n-1}}{(1+y)^{m-1}(1+y)^{n-1}(1+y)^2}\,dy$$

$$\beta(m,n)=\int_0^\infty \frac{y^{n-1}}{(1+y)^{m+n}}\,dy=\int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\,dx,$$

1.8.5. Relation between Gamma and Beta functions

Show that $\beta(m,n)=\frac{\Gamma m \Gamma n}{\Gamma(m+n)}, \qquad m>0,\ n>0$ **and also find the value of** $\beta\left(\frac{1}{2},\frac{1}{2}\right)$

(Nov- 2008, 2011)

Solution:

By definition of Gamma function $\Gamma(m) = \int_0^\infty e^{-x} x^{m-1} dx$

...(1)

Putting $x = y^2$ when $x = 0$, $y = 0$

$dx = 2ydy$ $x = \infty$, $y = \infty$

$$\Gamma(m) = \int_0^\infty e^{-y^2} (y^2)^{m-1} 2ydy$$

$$= 2\int_0^\infty e^{-y^2} y^{2m-1} dy \qquad ...(2)$$

Similarly $\Gamma(n) = 2\int_0^\infty e^{-x^2} x^{2n-1} dx$ ...(3)

Now from (2) and (3) we get,

$$\Gamma(m)\Gamma(n) = 4\int_0^\infty e^{-y^2} y^{2m-1} dy \int_0^\infty e^{-x^2} x^{2n-1} dx$$

$$= 4\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} x^{2n-1} y^{2m-1} dxdy \qquad ...(4)$$

Changing into to polar co-ordinates

$x = r\cos\theta$ $y = r\sin\theta$, $x^2 + y^2 = r^2$

$dxdy = rdrd\theta$

θ varies from 0 to $\frac{\pi}{2}$ and r varies from 0 to ∞

$$\Gamma(m)\Gamma(n) = 4\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} x^{2n-1} y^{2m-1} dxdy$$

$$= 4\int_0^{\frac{\pi}{2}}\int_0^{\infty} e^{-r^2}(r\cos\theta)^{2n-1}(r\sin\theta)^{2m-1} rdrd\theta$$

$$= 4\int_0^{\frac{\pi}{2}}\int_0^{\infty} e^{-r^2} r^{2n-1}\cos^{2n-1}\theta r^{2m-1}\sin^{2m-1}\theta rdrd\theta$$

$$= 4\int_0^{\frac{\pi}{2}}\cos^{2n-1}\theta\sin^{2m-1}\theta d\theta\int_0^{\infty} r^{2n+2m-2}e^{-r^2}\ rdrd\theta$$

$$= \beta(m,n)\ 2\int_0^{\infty} e^{-r^2} r^{2(m+n-1)} rdr \qquad \text{(By property 2)}$$

$$= \beta(m,n)\int_0^{\infty} e^{-t} t^{m+n-1} dt \qquad (\text{put } r^2 = t,\ 2rdr = dt\ )$$

$$\Gamma(m)\Gamma(n) = \beta(m,n)\Gamma(m+n)$$

$$\beta(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$$

To Find $\beta\left(\frac{1}{2},\frac{1}{2}\right)$

$$\beta\left(\frac{1}{2},\frac{1}{2}\right) = \frac{\Gamma\left(\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma(1)}$$

$$= \frac{\sqrt{\pi}\sqrt{\pi}}{1}$$

$$\beta\left(\frac{1}{2},\frac{1}{2}\right) = \pi$$

Important formulae:

1. Gamma function:

The Gamma Function (or) the Generalized Factorial Function denoted by $\Gamma(n)$ is defined as the definite integral $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx$ where $n > o$

2. Properties of Gamma function.

Property 1: $\Gamma(1) = 1$

Property 2: The reduction or recurrence formula is given by $\Gamma(n+1) = n\Gamma(n)$

Property 3: $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

Beta Function

1. The beta function which involves two parameters m and n is defined by the relation $\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1} dx, \quad m > 0, \quad n > 0$.

2. Another form of Beta function is $\beta(m,n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx, \quad m > 0, \quad n > 0$

3. Beta function in terms of circular functions is $\beta(m,n) = 2\int_0^{\frac{\pi}{2}} \sin^{2m-1}\theta \cos^{2n-1} d\theta$.

Note: The function $\beta(m,n)$ is symmetric**.** (i.e.) $\beta(m,n) = \beta(n,m)$**.**

Relation between Gamma and Beta functions: $\beta(m,n) = \frac{\Gamma m \Gamma n}{\Gamma(m+n)}$**.**

Example 1: Prove that $\int_0^\infty e^{-x^2}dx = \frac{\sqrt{\pi}}{2}$

Solution:

Put $x^2 = y$ when $x = 0, \quad y = 0$

$2xdx = dy \qquad x = \infty \quad y = \infty$

$$dx = \frac{dy}{2x}$$

$$dx = \frac{dy}{2\sqrt{y}}$$

$$\int_0^\infty e^{-x^2}dx = \int_0^\infty e^{-y}\frac{dy}{2\sqrt{y}}$$

$$= \frac{1}{2}\int_0^\infty e^{-y}y^{-1/2}dy = \frac{1}{2}\int_0^\infty e^{-x}x^{\frac{1}{2}-1}dx = \frac{1}{2}\Gamma\left(\frac{1}{2}\right)$$

$$= \frac{\sqrt{\pi}}{2}$$

$$\int_0^\infty e^{-x^2}dx = \frac{\sqrt{\pi}}{2}$$

Example 2: Show that $\int_0^{\frac{\pi}{2}} \sin^p\theta\cos^q\theta d\theta = \frac{1}{2}\frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{\Gamma\left(\frac{p+q+2}{2}\right)}$

Solution:

$$\beta(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)} \qquad ...(1)$$

$$\beta(m.n) = 2\int_0^{\frac{\pi}{2}} \sin^{2m-1}\theta \cos^{2n-1}\theta d\theta \qquad \ldots(2)$$

From (1) and (2) we get

$$2\int_0^{\frac{\pi}{2}} \sin^{2m-1}\theta \cos^{2n-1}\theta d\theta = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)} \qquad \ldots(3)$$

Putting $2m-1 = p \quad 2n-1 = q$

$$m = \frac{p+1}{2} \qquad n = \frac{q+1}{2}$$

Equation (3) becomes,

$$2\int_0^{\frac{\pi}{2}} \sin^{p}\theta \cos^{q}\theta d\theta = \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{\Gamma\left(\frac{p+q+2}{2}\right)}$$

$$\int_0^{\frac{\pi}{2}} \sin^{p}\theta \cos^{q}\theta d\theta = \frac{1}{2}\frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{\Gamma\left(\frac{p+q+2}{2}\right)} \qquad \ldots(4)$$

Put $p = n, \quad q = 0$ in (4), we get

$$\int_0^{\frac{\pi}{2}} \sin^{n}\theta d\theta = \frac{1}{2}\frac{\Gamma\left(\frac{n+1}{2}\right)\Gamma\left(1/2\right)}{\Gamma\left(\frac{n+2}{2}\right)}$$

$$\int_0^{\frac{\pi}{2}} \sin^{n}\theta d\theta = \frac{\sqrt{\pi}}{2}\frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n+2}{2}\right)}$$

Put $p = 0, \ q = n$ in (4) we get

$$\int_0^{\frac{\pi}{2}} \cos^n \theta d\theta = \frac{1}{2}\frac{\Gamma\left(\frac{1}{2}\right)\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n+2}{2}\right)} = \frac{\sqrt{\pi}}{2}\frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n+2}{2}\right)}$$

Example 3: Evaluate $\int_0^{\frac{\pi}{2}} \sin^{10} \theta d\theta$ **using Gamma function.**

Solution:

We know that

$$\int_0^{\frac{\pi}{2}} \sin^n \theta d\theta = \frac{1}{2}\frac{\Gamma\left(\frac{n+1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{n+2}{2}\right)}$$

Put n = 10

$$\int_0^{\frac{\pi}{2}} \sin^{10} \theta d\theta = \frac{1}{2}\frac{\Gamma\left(\frac{11}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{12}{2}\right)} \qquad \ldots(1)$$

We know that $\Gamma(n+1) = n\Gamma(n)$

$$\Gamma\left(\frac{11}{2}\right) = \Gamma\left(\frac{9}{2}+1\right) = \frac{9}{2}\Gamma\left(\frac{9}{2}\right)$$

$$= \frac{9}{2}\Gamma\left(\frac{7}{2}+1\right)$$

$$= \frac{9}{2}.\frac{7}{2}\Gamma\left(\frac{7}{2}\right)$$

$$= \frac{9}{2}.\frac{7}{2}\Gamma\left(\frac{5}{2}+1\right) \ldots.. = \frac{9}{2}.\frac{7}{2}.\frac{5}{2}.\frac{3}{2}.\frac{1}{2}\Gamma\left(\frac{1}{2}\right)$$

$$\Gamma\left(\frac{11}{2}\right)=\frac{63\times 15}{32}\sqrt{\pi}$$

Equation (1) becomes, $\int_0^{\frac{\pi}{2}}\sin^{10} d\theta=\frac{1}{2}\frac{63\times 15\times\sqrt{\pi}}{32}\times\frac{\sqrt{\pi}}{\Gamma(6)}$

$$=\frac{63\times 15}{2\times 32\times 120}\pi=\frac{63}{512}\pi \qquad [\because \Gamma(6)=5!=120]$$

Example 4: Evaluate $\int_0^{\frac{\pi}{2}}\cos^9\theta d\theta$ using Gamma functions

Solution:

We know that, $\int_0^{\frac{\pi}{2}}\cos^n\theta d\theta=\frac{1}{2}\frac{\Gamma\left(\frac{n+1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{n+2}{2}\right)}$

Put n = 9

$$\int_0^{\frac{\pi}{2}}\cos^9\theta d\theta=\frac{1}{2}\frac{\Gamma(5)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{11}{2}\right)}=\frac{1}{2}\frac{4!\sqrt{\pi}}{\Gamma\left(\frac{11}{2}\right)}$$

$$\Gamma\left(\frac{11}{2}\right)=\frac{63\times 15\times\sqrt{\pi}}{32} \qquad [\text{Refer example 3}]$$

$$\int_0^{\frac{\pi}{2}}\cos^9\theta d\theta=\frac{1}{2}\frac{24\sqrt{\pi}\times 32}{63\times 15\times\sqrt{\pi}}=\frac{128}{315}$$

Example 5: Evaluate $\int_0^{\frac{\pi}{2}}\sin^6\theta\cos^7\theta d\theta$ using Gamma function.

Solution:

Given that $\int_0^{\frac{\pi}{2}} \sin^6\theta\cos^7\theta d\theta$

We know that $\int_0^{\frac{\pi}{2}} \sin^p\theta\cos^q\theta d\theta = \frac{1}{2}\frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{\Gamma\left(\frac{p+q+2}{2}\right)}$ …(1)

Put $p=6, \quad q=7$ in (1)

$$\int_0^{\frac{\pi}{2}} \sin^6\theta\cos^7\theta d\theta = \frac{1}{2}\frac{\Gamma\left(\frac{7}{2}\right)\Gamma\left(\frac{8}{2}\right)}{\Gamma\left(\frac{15}{2}\right)}$$

$$= \frac{1}{2}\frac{\Gamma\left(\frac{7}{2}\right)\Gamma(4)}{\Gamma\left(\frac{15}{2}\right)} \quad = \frac{1}{2}\frac{6\sqrt{\pi}\times 16}{\sqrt{\pi}13\times 11\times 9\times 7}$$

$$\int_0^{\frac{\pi}{2}} \sin^6\theta\cos^7\theta d\theta = \frac{16}{3003}$$

Example 6: Evaluate $\int_0^{\frac{\pi}{2}} \sin^{\frac{7}{2}}\theta\cos^{\frac{5}{2}}\theta d\theta$

Solution:

We know that $\int_0^{\frac{\pi}{2}} \sin^p\theta\cos^q\theta d\theta = \frac{1}{2}\frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{\Gamma\left(\frac{p+q+2}{2}\right)}$ …(1)

Put $p=\frac{7}{2}, \quad q=\frac{5}{2}$ in (1)

$$\int_0^{\frac{\pi}{2}} \sin^{7/2}\theta \cos^{5/2}\theta d\theta = \frac{1}{2}\frac{\Gamma\left(\frac{\frac{7}{2}+1}{2}\right)\Gamma\left(\frac{\frac{5}{2}+1}{2}\right)}{\Gamma\left(\frac{\frac{7}{2}+\frac{5}{2}+2}{2}\right)}$$

$$= \frac{1}{2}\frac{\Gamma\left(\frac{9}{4}\right)\Gamma\left(\frac{7}{4}\right)}{\Gamma 4} \qquad = \frac{1}{2}\frac{\frac{5}{4}.\frac{1}{4}\Gamma\left(\frac{1}{4}\right).\frac{3}{4}.\Gamma\left(\frac{3}{4}\right)}{3!}$$

$$= \frac{15}{2\times 64\times 6}\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{3}{4}\right)$$

$$\int_0^{\frac{\pi}{2}} \sin^{\frac{7}{2}}\theta \cos^{\frac{5}{2}}\theta d\theta = \frac{5}{256}\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{3}{4}\right)$$

Example 7: Show that $\int_0^{\frac{\pi}{2}} \sqrt{\tan\theta}\, d\theta = \frac{\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{3}{4}\right)}{2}$ **(Nov- 2000, May- 2001)**

Solution:

We know that $\int_0^{\frac{\pi}{2}} \sqrt{\tan\theta}\, d\theta = \int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin\theta}}{\sqrt{\cos\theta}} d\theta$

$$= \int_0^{\frac{\pi}{2}} \sin^{\frac{1}{2}}\theta \cos^{\frac{-1}{2}}\theta d\theta$$

We know that $\int_0^{\frac{\pi}{2}} \sin^p\theta \cos^q d\theta = \frac{1}{2}\frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{\Gamma\left(\frac{p+q+2}{2}\right)}$

Put $p=\frac{1}{2}, \quad q=-\frac{1}{2}$ in we get,

$$\int_0^{\frac{\pi}{2}} \sin^{\frac{1}{2}}\theta\cos^{\frac{1}{2}}d\theta=\frac{1}{2}\frac{\Gamma\left(\frac{\frac{1}{2}+1}{2}\right)\Gamma\left(\frac{-\frac{1}{2}+1}{2}\right)}{\Gamma\left(\frac{\frac{1}{2}-\frac{1}{2}+2}{2}\right)}$$

$$=\frac{1}{2}\frac{\Gamma\left(\frac{3}{4}\right)\Gamma\left(\frac{1}{4}\right)}{\Gamma(1)}=\frac{\Gamma\left(\frac{3}{4}\right)\Gamma\left(\frac{1}{4}\right)}{2}$$

Example 8: Show that $\int_0^1 \frac{x^n}{\sqrt{1-x^2}}dx=\frac{1.3.5.....(n-1)}{2.4.6.......n}\frac{1}{2}\pi$ **if n is an even integer**

(Nov-2006)

Solution:

Let $x=\sin\theta,$

$dx=\cos\theta d\theta$

When $x=1, \ \theta=\frac{\pi}{2}$

$x=0, \quad \theta=0$

$$\int_0^1 \frac{x^n}{\sqrt{1-x^2}}dx=\int_0^{\frac{\pi}{2}}\frac{\sin^n\theta\cos\theta d\theta}{\sqrt{1-\sin^2\theta}}$$

$$=\int_0^{\frac{\pi}{2}}\frac{\sin^n\theta\cos\theta d\theta}{\cos\theta}=\int_0^{\frac{\pi}{2}}\sin^n\theta d\theta$$

$$= \frac{1}{2} \frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n+2}{2}\right)} \Gamma\left(\frac{1}{2}\right) \qquad ...(1)$$

We know that $\Gamma(n+1) = n\Gamma n$

$$\Gamma\left(\frac{n+1}{2}\right) = \left(\frac{n+1}{2} - 1\right)\Gamma\left(\frac{n+1}{2} - 1\right)$$

$$= \left(\frac{n-1}{2}\right)\Gamma\left(\frac{n-1}{2}\right) \qquad ...(2)$$

$$\Gamma\left(\frac{n-1}{2}\right) = \left(\frac{n-1}{2} - 1\right)\Gamma\left(\frac{n-1}{2} - 1\right)$$

$$= \left(\frac{n-3}{2}\right)\Gamma\left(\frac{n-3}{2}\right) \qquad ...(3)$$

$$\therefore \Gamma\left(\frac{n+1}{2}\right) = \left(\frac{n-1}{2}\right)\left(\frac{n-3}{2}\right)\Gamma\left(\frac{n-3}{2}\right)$$

$$= \left(\frac{n-1}{2}\right)\left(\frac{n-3}{2}\right)\left(\frac{n-5}{2}\right)........\frac{1}{2}\Gamma\left(\frac{1}{2}\right) \qquad ...(4)$$

Changing n into n+1 in (4)

$$\Gamma\left(\frac{n+2}{2}\right) = \frac{n}{2}.\frac{n-2}{2}.\frac{n-4}{2}...\frac{2}{2}\Gamma\left(\frac{2}{2}\right) \qquad ...(5)$$

$$\frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n+2}{2}\right)} = \frac{(n-1)(n-3)(n-5)...\Gamma\left(\frac{1}{2}\right)}{n(n-2)(n-4)...2\Gamma(1)} \qquad ...(6)$$

Substituting (6) in (1), we get

$$\int_0^1 \frac{x^n}{\sqrt{1-x^2}}\,dx = \frac{1}{2}\frac{(n-1)(n-3)(n-5)...1\Gamma\left(\frac{1}{2}\right)}{n(n-2)(n-4)...2\Gamma(1)}\Gamma\left(\frac{1}{2}\right)$$

$$= \frac{1}{2}\frac{1.3.5...(n-1)\sqrt{\pi}}{2.4.6...n}\sqrt{\pi}$$

$$= \frac{1.3.5...(n-1)}{2.4.6...n}\frac{\pi}{2}$$

Example 9: Express $\int_0^1 x^m (1-x^n)^p \, dx$ in terms of gamma functions and evaluate

$\int_0^1 x^5 (1-x^3)^{10} \, dx$ **(Apr- 2003, 2008, 2010)**

Solution:

Let $I = \int_0^1 x^m (1-x^n)^p \, dx$

Put $x^n = u$

$nx^{n-1} dx = du$

When $x = 0, \quad u = 0$

$x = 1, \quad u = 1$

$$I = \int_0^1 u^{\frac{m}{n}} (1-u)^p \frac{du}{nu^{\frac{n-1}{n}}} = \frac{1}{n}\int_0^1 u^{\frac{m}{n}-\frac{n-1}{n}} (1-u)^p \, du$$

$$= \frac{1}{n}\int_0^1 u^{\frac{m-n+1}{n}} (1-u)^p \, du$$

$$= \frac{1}{n}\int_0^1 u^{\frac{m+1}{n}-1} (1-u)^{p+1-1} \, du$$

$$= \frac{1}{n}\beta\left[\frac{m+1}{n}, p+1\right] \qquad \left(\because \beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1} dx\right)$$

To Find $\int_0^1 x^5 (1-x^3)^{10} \, dx$

$$\int_0^1 x^m (1-x^n)^p \, dx = \frac{1}{n} \frac{\Gamma\left(\frac{m+1}{n}\right)\Gamma(p+1)}{\Gamma\left(\frac{m+1}{n}+p+1\right)} \qquad \left(\because \beta(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}\right)$$

put m = 5 ,n = 3 and p = 10 we get,

$$\int_0^1 x^5 (1-x^3)^{10} \, dx = \frac{1}{3} \frac{\Gamma\left(\frac{5+1}{3}\right)\Gamma(10+1)}{\Gamma\left(\frac{5+1}{3}+10+1\right)}$$

$$= \frac{1}{3} \frac{\Gamma(2)\Gamma(11)}{\Gamma(13)}$$

$$= \frac{1}{396}$$

Example 10. Show that $\beta(m,n) = \int_0^\infty \frac{y^{n-1}}{(1+y)^{m+n}} dy = \int_0^1 \frac{x^{n-1}+x^{m-1}}{(1+x)^{m+n}} dx$

Solution:

We prove the result $\beta(m,n) = \int_0^\infty \frac{y^{n-1}}{(1+y)^{m+n}} dy$

Now $\beta(m,n) = \int_0^\infty \frac{y^{n-1}}{(1+y)^{m+n}} dy$

$$= \int_0^1 \frac{y^{n-1}}{(1+y)^{m+n}} dy + \int_1^\infty \frac{y^{n-1}}{(1+y)^{m+n}} dy \qquad \ldots(1)$$

Consider the integral $\int_1^\infty \frac{y^{n-1}}{(1+y)^{m+n}} dy$

Put $y = \frac{1}{z}$

$dy = -\frac{1}{z^2}\, dz$

When $y = \infty, \quad z = 0$

$y = 1, \quad$ z =1

$$\int_1^\infty \frac{y^{n-1}}{(1+y)^{m+n}}\, dy = \int_1^0 \frac{\left(\frac{1}{z}\right)^{n-1}}{\left(1+\frac{1}{z}\right)^{m+n}} \left(-\frac{1}{z^2}\, dz\right)$$

$$= \int_0^1 \frac{\left(\frac{1}{z^{n-1}}\right)}{\left(\frac{z+1}{z}\right)^{m+n}} \left(\frac{1}{z^2}\, dz\right)$$

$$= \int_0^1 \frac{z^{m+n}}{z^{n-1}\left(1+z\right)^{m+n} z^2} dz$$

$$= \int_0^1 \frac{z^{m+n}}{z^{n+1}\left(1+z\right)^{m+n}} dz$$

$$= \int_0^1 \frac{z^{m-1}}{\left(1+z\right)^{m+n}} dz$$

$$\beta(m,n) = \int_0^\infty \frac{y^{n-1}}{(1+y)^{m+n}}\, dy + \int_0^1 \frac{z^{m-1}}{\left(1+z\right)^{m+n}} dz$$

$$= \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}}\, dy + \int_0^1 \frac{x^{m-1}}{\left(1+x\right)^{m+n}} dz$$

$$\beta(m,n) = \int_0^1 \frac{x^{n-1} + x^{m-1}}{(1+x)^{m+n}}\, dx$$

Example 11: Evaluate $\int_0^1 x^m \left(\log \frac{1}{x}\right)^n dx$ **.Hence find the value of** $\int_0^1 x^4 \left(\log \frac{1}{x}\right)^3 dx$

(Nov- 2001,2003,2004)

Solution:

Put $\log\left(\frac{1}{x}\right) = t$

$\frac{1}{x} = e^{t}$

$x = e^{-t}$ when x = 0, t = ∞

$dx = -e^{-t}dt$ x = 1, t =1

$$\int_0^1 x^m \left(\log\frac{1}{x}\right)^n dx = \int_\infty^0 \left(e^{-t}\right)^m t^n \left(-e^{-t}\right)dt$$

$$= \int_0^\infty e^{-mt}e^{-t}t^n dt$$

$$= \int_\infty^0 e^{-(m+1)t}t^n dt \qquad \ldots(1)$$

Put $(1+m)t = u$ when t = 0, u = 0

$(1+m)dt = du$ t =∞, u = ∞

$$\int_0^\infty e^{-(m+1)t}t^n dt = \int_\infty^0 e^{-u}\left(\frac{u}{1+m}\right)^n \frac{du}{1+m}$$

$$= \frac{1}{(1+m)^{n+1}}\int_\infty^0 e^{-u}u^n du$$

$$= \frac{\Gamma(n+1)}{(1+m)^{n+1}} = \frac{n!}{(1+m)^{n+1}} \qquad \ldots(2)$$

To evaluate $\int_0^1 x^4\left(\log\frac{1}{x}\right)^3 dx$

put m = 4, n = 3, in (2) we get

$$\int_0^1 x^4\left(\log\frac{1}{x}\right)^3 dx=\frac{3!}{(1+4)^{3+1}}=\frac{6}{625}$$

Example 12: Show that $\int_0^1 x^m(\log x)^n\,dx=\frac{(-1)^n\,n!}{(m+1)^{n+1}}$ **, n is an integer**

Solution:

Put $\log x=t$ when x = 0, t = ∞

$x=e^t$ x =1, t = 0

$dx=e^t\mathrm{dt}$ $(\log 0=-\infty,\log 1=0)$

$$\int_0^1 x^m(\log x)^n\,dx=\int_{-\infty}^0 e^m t^n e^t\,dt$$

$$=\int_{-\infty}^0 e^{(1+m)t}t^n\,dt$$

Put $(1+m)t=-u$ when $t=-\infty,\ u=\infty$

$(1+m)dt=-du$ $t=0,\ u=0$

$$\int_0^1 x^m(\log x)^n\,dx=\int_{-\infty}^0 e^{(1+m)t}t^n\,dt$$

$$=\int_\infty^0 e^{-u}\left(\frac{-u}{1+m}\right)^n\left(\frac{-du}{1+m}\right)$$

$$=\frac{1}{(1+m)^{n+1}}\int_\infty^0 e^{-u}(-1)^n u^n(-du)$$

$$=\frac{(-1)^n}{(1+m)^{n+1}}\int_0^\infty e^{-u}u^n\,du$$

$$=\frac{(-1)^n}{(1+m)^{n+1}}\Gamma(n+1)=\frac{(-1)^n\,n!}{(m+1)^{n+1}}$$

Example 13: Evaluate $\int_0^{\infty} \frac{dx}{x^4+1}$

Solution: Given that $\int_0^{\infty} \frac{dx}{x^4+1}$

Put $x^2 = \tan\theta$ when $x = 0,\ \theta = 0$

$2x\,dx = \sec^2\theta\,d\theta$ $x = \infty,\ \theta = \frac{\pi}{2}$

$$dx = \frac{\sec^2\theta}{2x}\,d\theta$$

$$\int_0^{\infty} \frac{dx}{x^4+1} = \int_0^{\frac{\pi}{2}} \frac{\sec^2\theta}{\left(\tan^2\theta + 1\right)^2 2\sqrt{\tan\theta}}\,d\theta$$

$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} \frac{\sec^2\theta}{\sec^4\theta(\tan\theta)^{\frac{1}{2}}}\,d\theta$$

$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} \cos^{\frac{1}{2}}\theta\sin^{-\frac{1}{2}}\,d\theta = \frac{1}{4}\frac{\Gamma\left(\frac{-\frac{1}{2}+1}{2}\right)\Gamma\left(\frac{\frac{1}{2}+1}{2}\right)}{\Gamma\left(\frac{-\frac{1}{2}+\frac{1}{2}+2}{2}\right)}$$

$$= \frac{1}{4}\frac{\Gamma\left(\frac{3}{4}\right)\Gamma\left(\frac{1}{4}\right)}{\Gamma(1)}$$

$$\int_0^{\infty} \frac{dx}{x^4+1} = \frac{1}{4}\Gamma\left(\frac{3}{4}\right)\Gamma\left(\frac{1}{4}\right)$$

Example 14: Show that $\beta\left(m,\frac{1}{2}\right)=2^{2m-1}\beta(m,m)$ **(Nov -2004)**

Solution:

We know that $\beta(m,n)=2\int_0^{\frac{\pi}{2}}\sin^{2m-1}\theta\cos^{2n-1}\theta\,d\theta$ … (1)

Put $n=\frac{1}{2}$ in (1)

$$\beta\left(m,\frac{1}{2}\right)=2\int_0^{\frac{\pi}{2}}\sin^{2m-1}\theta\,d\theta \quad \text{… (2)}$$

Put $n=m$ in (1)

$$\beta(m,m)=2\int_0^{\frac{\pi}{2}}\sin^{2m-1}\theta\cos^{2m-1}\theta\,d\theta$$

$$=2\int_0^{\frac{\pi}{2}}(\sin\theta\cos\theta)^{2m-1}\,d\theta$$

$$=2\int_0^{\frac{\pi}{2}}\left(\frac{\sin 2\theta}{2}\right)^{2m-1}d\theta$$

put $2\theta=t$

$2d\theta=dt$

$d\theta=\frac{dt}{2}$

when $\theta=0,\ t=0$

$\theta=\frac{\pi}{2},\ t=\pi$

$$=2\int_0^{\pi}\frac{1}{2^{2m-1}}\sin^{2m-1}t\,\frac{dt}{2} = \frac{1}{2^{2m-1}}\int_0^{\pi}\sin^{2m-1}t\,dt$$

$$\beta(m,m)=\frac{1}{2^{2m-1}}2\int_0^{\frac{\pi}{2}}\sin^{2m-1}t\,dt$$

$$2^{2m-1}\beta(m,m)=2\int_0^{\frac{\pi}{2}}\sin^{2m-1}t\,dt$$

$$=2\int_0^{\frac{\pi}{2}}\sin^{2m-1}\theta\,d\theta=\beta\left(m,\frac{1}{2}\right)\qquad\text{(by (2))}$$

$$\beta\left(m,\frac{1}{2}\right)=2^{2m-1}\beta(m,m)$$

Example 15: Prove that the duplication formula $\Gamma(2m)=\frac{2^{2m-1}}{\sqrt{\pi}}\Gamma(m)\Gamma\left(m+\frac{1}{2}\right)$

(Nov -2008)

Solution:

We know that $\beta\left(m,\frac{1}{2}\right)=2^{2m-1}\beta(m,m)$

$$\frac{\Gamma(m)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(m+\frac{1}{2}\right)}=\frac{2^{2m-1}\Gamma(m)\Gamma(m)}{\Gamma(m+m)}\qquad\left[\therefore\beta(m,n)=\frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}\right]$$

$$\frac{\Gamma(m)\sqrt{\pi}}{\Gamma\left(m+\frac{1}{2}\right)}=\frac{2^{2m-1}\Gamma(m)\Gamma(m)}{\Gamma(2m)}$$

$$\Gamma(m)\Gamma(2m)=\frac{2^{2m-1}(\Gamma(m))^2\,\Gamma\left(m+\frac{1}{2}\right)}{\sqrt{\pi}}$$

$$\Gamma(2m)=\frac{2^{2m-1}\Gamma(m)\Gamma\left(m+\frac{1}{2}\right)}{\sqrt{\pi}}$$

Hence proved.

Example 16: Prove that $\dfrac{\beta(m+1,n)}{m}=\dfrac{\beta(m,n+1)}{n}=\dfrac{\beta(m,n)}{m+n}$

(May- 2007, Nov- 2007)

Solution:

Consider $\dfrac{\beta(m+1,n)}{m}=\dfrac{\Gamma(m+1)\Gamma(n)}{m\,\Gamma(m+1+n)}$ $\qquad \left[\therefore \beta(m,n)=\dfrac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}\right]$

$$= \frac{m\ \Gamma(m)\ \Gamma(n)}{m(m+n)\Gamma(m+n)} \qquad \Gamma(n+1)=n\Gamma(n)$$

$$= \frac{1}{m+n}\,\frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$$

$$= \frac{\beta(m,n)}{m+n} \qquad \ldots (1)$$

Consider $\dfrac{\beta(m,n+1)}{n}=\dfrac{\Gamma(m)\ \Gamma(n+1)}{\Gamma(m+n+1)\cdot n}$

$$= \frac{\Gamma(m)\ n\Gamma(n)}{n(m+n)\Gamma(m+n)}$$

$$= \frac{\Gamma(m)\ \Gamma(n)}{(m+n)\Gamma(m+n)}$$

$$= \frac{\beta(m,n)}{(m+n)} \qquad \ldots (2)$$

From (1) and (2) we get

$$\frac{\beta(m+1,n)}{m} = \frac{\beta(m,n+1)}{n} = \frac{\beta(m,n)}{m+n}$$

Example 17: Prove that $\beta(m,n)=\dfrac{\sqrt{\pi}\,\Gamma(n)}{2^{2n-1}\Gamma\left(n+\frac{1}{2}\right)}$ **(April- 2003)**

Solution:

By definition, $\beta\left(m,n\right) = 2\int_0^{\frac{\pi}{2}} \sin^{2n-1}\theta \cos^{2n-1}\theta \, d\theta$

$$= 2\int_0^{\frac{\pi}{2}} \left(\frac{\sin 2\theta}{2}\right)^{2n-1} d\theta$$

$$= \frac{1}{2^{2n-2}} \int_0^{\frac{\pi}{2}} \sin^{2n-1} 2\theta \, d\theta$$

Put $2\theta = \varphi$ when $\theta = 0$, $\varphi = 0$

$2d\theta = d\varphi$ $\theta = \frac{\pi}{2}$, $\varphi = \pi$

$d\theta = \frac{d\varphi}{2}$

$$= \frac{1}{2^{2n-2}} 2\int_0^{\frac{\pi}{2}} \sin^{2n-1}\varphi \frac{d\varphi}{2} = \frac{1}{2^{2n-2}} \int_0^{\frac{\pi}{2}} \sin^{2n-1}\varphi \, d\varphi$$

$$= \frac{1}{2^{2n-2}} \frac{1}{2} \beta\left(n, \frac{1}{2}\right) = \frac{1}{2^{2n-1}} \beta\left(n, \frac{1}{2}\right)$$

$$= \frac{1}{2^{2n-1}} \frac{\Gamma(n)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(n+\frac{1}{2}\right)} = \frac{1}{2^{2n-1}} \frac{\Gamma(n)\sqrt{\pi}}{\Gamma\left(n+\frac{1}{2}\right)}$$

Example 18: Show that $\int_0^1 \frac{x^{2a}}{\sqrt{1-x^2}} dx = \frac{\sqrt{\pi}}{2} \frac{\Gamma\left(a+\frac{1}{2}\right)}{\Gamma\left(a+1\right)}$

Solution:

Consider L.H.S. = $\int_0^1 \frac{x^{2a}}{\sqrt{1-x^2}} dx$

Put $x=\sin\theta$ when $x=0$, $\theta=0$

$dx=\cos\theta\, d\theta$ $\quad x=1$, $\theta=\frac{\pi}{2}$

$x^{2a}=\sin^{2a}\theta$

$$\int_0^1 \frac{x^{2a}}{\sqrt{1-x^2}}dx=\int_0^{\frac{\pi}{2}}\frac{\sin^{2a}\theta}{\sqrt{1-\sin^2\theta}}\cos\theta\, d\theta$$

$$=\int_0^{\frac{\pi}{2}}\frac{\sin^{2a}\theta}{\cos\theta}\cos\theta\, d\theta$$

$$=\int_0^{\frac{\pi}{2}}\sin^{2a}\theta\, d\theta$$

$$\int_0^{\frac{\pi}{2}}\sin^n\theta\, d\theta=\frac{\sqrt{\pi}}{2}\frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n+2}{2}\right)}$$

$$=\frac{\sqrt{\pi}}{2}\frac{\Gamma\left(\frac{2a+1}{2}\right)}{\Gamma\left(\frac{2a+2}{2}\right)}$$

$$\int_0^1 \frac{x^{2a}}{\sqrt{1-x^2}}dx=\frac{\sqrt{\pi}}{2}\frac{\Gamma\left(a+\frac{1}{2}\right)}{\Gamma(a+1)}$$

Example 19: Evaluate $\int_0^1 \frac{dx}{\sqrt[n]{1-x^n}}$

Solution:

Given that $\int_0^1 \frac{dx}{\sqrt[n]{1-x^n}}$

Let $x^n=\sin^2\theta$, $x=\sin^{\frac{2}{n}}\theta$

$nx^{n-1}dx=2\sin\theta\cos\theta\, d\theta$

$$dx = \frac{2\sin\theta\cos\theta}{nx^{n-1}}d\theta$$

$$dx = \frac{2\sin\theta\cos\theta}{n\left(\sin^{2/n}\theta\right)^{n-1}}d\theta$$

$$dx = \frac{2\sin\theta\cos\theta}{n\sin^{\frac{2n-2}{n}}\theta}d\theta$$

When $x = 0$, $\theta = 0$

$$x = 1,\ \theta = \frac{\pi}{2}$$

$$\int_0^1 \frac{dx}{\sqrt[n]{1-x^n}} = \int_0^{\frac{\pi}{2}} \frac{\frac{2\sin\theta\cos\theta}{n\sin^{\frac{2n-2}{n}}\theta}}{\sqrt[n]{1-\sin^2\theta}}d\theta$$

$$= \int_0^{\frac{\pi}{2}} \frac{\left(\frac{2\sin\theta\cos\theta}{n\sin^{2-\frac{2}{n}}\theta}\right)}{\cos^{\frac{2}{n}}\theta}d\theta$$

$$= \frac{2}{n}\int_0^{\frac{\pi}{2}} \sin^{\left(1-2+\frac{2}{n}\right)}\theta\cos^{\left(1-\frac{2}{n}\right)}\theta\, d\theta$$

$$= \frac{2}{n}\int_0^{\frac{\pi}{2}} \sin^{\left(\frac{2}{n}-1\right)}\theta\cos^{\left(1-\frac{2}{n}\right)}\theta\, d\theta$$

$$= \frac{2}{n}\frac{\Gamma\left(\frac{\left(\frac{2}{n}-1+1\right)}{2}\right)\Gamma\left(\frac{\left(1-\frac{2}{n}+1\right)}{2}\right)}{\Gamma\left(\frac{\frac{2}{n}-1+1+2-\frac{2}{n}}{2}\right)}$$

$$= \frac{2}{n}\frac{\Gamma\left(\frac{1}{n}\right)\Gamma\left(\frac{n-1}{n}\right)}{\Gamma(1)}$$

Example 20: Evaluate $\int_0^1 \frac{dx}{\sqrt{1-x^4}}$

Solution:

Given that $\int_0^1 \frac{dx}{\sqrt{1-x^4}}$

Put $x^4 = \sin^2\theta$

$x = \sqrt{\sin\theta}$ when $x = 0,\ \theta = 0$

$dx = \frac{1}{2}\sin^{\frac{1}{2}-1}\theta\cos\theta\, d\theta$ $\qquad x = 1,\ \theta = \frac{\pi}{2}$

$$\int_0^1 \frac{dx}{\sqrt{1-x^4}} = \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1-\sin^2\theta}}\frac{1}{2}\sin^{-\frac{1}{2}}\theta\cos\theta\, d\theta$$

$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} \frac{\sin^{-\frac{1}{2}}\theta}{\cos\theta}\cos\theta\ d\theta = \frac{1}{2}\int_0^{\frac{\pi}{2}} \sin^{-\frac{1}{2}}\theta\, d\theta$$

$$= \frac{1}{2}\frac{\sqrt{\pi}}{2}\frac{\Gamma\left(\frac{\frac{-1}{2}+1}{2}\right)}{\Gamma\left(\frac{\frac{-1}{2}+2}{2}\right)} = \frac{\sqrt{\pi}}{4}\ \frac{\Gamma\left(\frac{1}{4}\right)}{\Gamma\left(\frac{3}{4}\right)}$$

$$\int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{\sqrt{\pi}\ \Gamma\left(\frac{1}{4}\right)}{4\Gamma\left(\frac{3}{4}\right)}$$

Example 21: Evaluate $\int_0^\infty e^{-x}x^4\,dx$ **.**

Solution:

Consider $\int_0^\infty e^{-x}x^{n-1}\,dx = \Gamma(n)$

$$\int_0^\infty e^{-x}x^4\,dx = \int_0^\infty e^{-x}x^{5-1}\,dx$$

$$=\Gamma(5)$$

$$= \Gamma(4+1) = 4! = 24.$$

Example 22: Evaluate $\int_0^\infty e^{-x^2}x^4\,dx$

Solution:

Consider $\int_0^\infty e^{-x^2}x^4\,dx$

Put $x^2 = t$ When $x = 0,\ t = 0$

$2x\,dx = dt$ $x = \infty,\ t = \infty$

$$\int_0^\infty e^{-x^2}x^4\,dx = \int_0^\infty e^{-x^2}x^3\,xdx$$

$$= \int_0^\infty e^{-t}t\sqrt{t}\,\frac{dt}{2}$$

$$= \frac{1}{2}\int_0^\infty e^{-t}t^{\frac{3}{2}}\,dt$$

$$= \frac{1}{2}\int_0^\infty e^{-t}t^{\frac{5}{2}-1}\,dt \qquad = \frac{1}{2}\Gamma\left(\frac{5}{2}\right)$$

$$= \frac{1}{2}\Gamma\left(\frac{3}{2}+1\right) = \frac{1}{2}\ \ \frac{3}{2}\Gamma\left(\frac{3}{2}\right)$$

$$= \frac{1}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\Gamma\left(\frac{1}{2}\right)$$

$$= \frac{3}{8}\sqrt{\pi}$$

Example 23: Prove that $\beta(m,m)\cdot\beta\left(m+\frac{1}{2},m+\frac{1}{2}\right)=\pi m^{-1}2^{1-4m}$

Solution:

Consider L.H.S= $\beta(m,m) = \frac{2}{2^{2m-1}}\int_0^{\frac{1}{2}\pi}\sin^{2m-1}\alpha d\alpha$ … (1)

$$\beta\left(m+\frac{1}{2},m+\frac{1}{2}\right) = \frac{2}{2^{2\left(m+\frac{1}{2}\right)-1}}\int_0^{\frac{\pi}{2}}\sin^{2\left(m+\frac{1}{2}\right)-1}\alpha\ d\alpha$$

$$= \frac{2}{2^{2m}}\int_0^{\frac{\pi}{2}}\sin^{2m}\alpha\ d\alpha \qquad \dots (2)$$

Multiplying (1) and (2), we have,

$$\beta(m,m)\cdot\beta\left(m+\frac{1}{2},m+\frac{1}{2}\right) = \frac{2}{2^{2m-1}}\int_0^{\frac{\pi}{2}}\sin^{2m-1}\alpha d\alpha\times\frac{2}{2^{2m}}\int_0^{\frac{\pi}{2}}\sin^{2m}\alpha d\alpha$$

$$= \frac{4}{2^{2m-1+2m}}\int_0^{\frac{\pi}{2}}\sin^{2m-1}\alpha d\alpha\int_0^{\frac{\pi}{2}}\sin^{2m}\alpha d\alpha$$

$$= \frac{4}{2^{4m-1}}\frac{2\cdot4\cdot6\cdots(2m-2)}{1\cdot3\cdot5\cdots(2m-1)}\frac{1\cdot3\cdot5\cdots(2m-1)}{2\cdot4\cdot6\cdots2m}\frac{\pi}{2}$$

$$\left(\because \int_0^{\frac{\pi}{2}} \sin^{2m-1}\alpha\, d\alpha = \frac{2\cdot 4\cdot 6\cdots(2m-2)}{1\cdot 3\cdot 5\cdots(2m-1)} \quad \int_0^{\frac{\pi}{2}} \sin^{2m}\alpha\, d\alpha = \frac{1\cdot 3\cdot 5\cdots(2m-1)}{2\cdot 4\cdot 6\cdots 2m}\frac{\pi}{2}\right)$$

$$= \frac{4}{2^{4m-1}}\frac{1}{2m}\frac{\pi}{2} = \pi m^{-1}\, 2^{1-4m}$$

Example 24: Show that $\frac{d^n}{dy^n}\Gamma(y) = \int_0^\infty x^{y-1}e^{-x}(\log x)^n\, dx$**.**

Solution:

$$\text{Consider } \Gamma(y) = \int_0^\infty e^{-x}x^{y-1}dx$$

$$= \int_0^\infty e^{-x}\frac{x^y}{x}dx$$

$$\frac{d^n}{dy^n}\Gamma\left(y\right) = \frac{d^n}{dy^n}\int_0^\infty \frac{e^{-x}}{x}x^y dx$$

$$= \int_0^\infty \frac{e^{-x}}{x}\frac{d^n}{dy^n}\left(x^y\right)dx$$

$$\frac{d}{dx}\left(a^x\right) = a^x \log a$$

$$\frac{d^n}{dx^n}\left(a^x\right) = a^x\left(\log a\right)^n$$

$$= \int_0^\infty \frac{e^{-x}}{x}x^y\left(\log x\right)^n dx$$

$$= \int_0^\infty x^{y-1}e^{-x}\left(\log x\right)^n dx.$$

Exercise

1. Express the following integrals in terms of Gamma functions

(i) $\frac{1}{n}\int_0^\infty e^{-x^3}dx$ (ii) $\int_0^\infty x^6 e^{-3x}dx$ (iii) $\int_0^\infty e^{-x^2}dx$ (iv) $\int_0^\infty x^n e^{-ax}dx$ (v) $\int_0^\infty x^m\left(\log\frac{1}{n}\right)^n dx$

(vi) $\int_0^1 x^4\left(\log\frac{1}{n}\right)^3 dx$ (vii) $\int_0^a \sqrt{a^2-x^2}\,x^4 dx$

2. Show that $\int_0^{\frac{\pi}{2}}\sqrt{\sin\theta}\,d\theta x\int_0^{\frac{\pi}{2}}\frac{1}{\sqrt{\sin\theta}}d\theta=\pi$

3. Show that $\int_0^1\frac{dx}{\sqrt{1-x^n}}=\frac{\sqrt{\pi}}{n}\frac{\Gamma\frac{1}{n}}{\Gamma\left(\frac{1}{n}+\frac{1}{2}\right)}$

4. Show that $\Gamma\left(\frac{2k+1}{k}\right)=\frac{1.3.5.....(2k-1)}{2k}\sqrt{\pi}$

5. Show that $\int_0^\infty x^{n-1}e^{-ax}dx=a^{-n}\Gamma n$ if $a>o$

6. Show that $\int_0^1\frac{x^{2a}}{\sqrt{1-x^2}}dx=\frac{\sqrt{\pi}}{2}\frac{\Gamma\left(a+\frac{1}{2}\right)}{\Gamma(a+1)}$.

7. Show that $\int_0^\infty x^{n-1}e^{-ax}dx=a^{-n}\Gamma(n), a>0$.

8. Show that $\Gamma(p)\Gamma(1-p)=\frac{\pi}{\sin p\pi}$.

9. Evaluate $\int_0^\infty e^{-x^2}dx$.

10. Evaluate $\int_0^1\frac{dx}{\sqrt{1-x^4}}dx$.

11. Evaluate $\int_0^{\frac{\pi}{2}}\sin^5\theta\cos^6\theta\,d\theta$.

12. Evaluate $\int_0^1 x^7 (1-x)^8 dx$.

13. Prove that $\int_0^\infty \sqrt{x}e^{-x^2}dx \times \int_0^\infty \frac{e^{-x^2}}{\sqrt{x}}dx = \frac{\pi}{2\sqrt{2}}$.

14. Prove that $\int_0^\infty \frac{x^2dx}{\sqrt{1-x^4}} \times \int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{\pi}{4}$

15. Prove that $\beta(m+1,n)+\beta(m,n+1)=\beta(m,n)$

16. Find the value of $\iint x^m y^n dxdy,$ taken over the area $x \geq 0$, $x+y \leq 1$ in terms of gamma fn, if $m,n>0$

Answers

1.(i) $\frac{1}{3}\Gamma\left(\frac{1}{3}\right)$ (ii) $\frac{1}{3^7}\Gamma(7)$ (iii) $\frac{\sqrt{\pi}}{2}$ (iv) $\frac{\Gamma(n+1)}{a^{n+1}}$

(v) $\frac{1}{(m+1)^{n+1}}\Gamma(n+1)$ (vi) $\frac{1}{54}\Gamma(14)$ (vii) $\frac{a^6}{2}\frac{\Gamma\left(\frac{5}{2}\right)\Gamma\left(\frac{3}{2}\right)}{\Gamma(4)}$

9. $\frac{\sqrt{\pi}}{2}$

10. $\frac{\sqrt{\pi}}{4}\frac{\Gamma\left(\frac{1}{4}\right)}{\Gamma\left(\frac{3}{4}\right)}$

11. $\frac{16}{693}$

12. $\frac{63\pi}{512}$

13. $\frac{7!\times 8!}{16!}$

14. $\frac{\Gamma(m+1)\Gamma(n+1)}{\Gamma(m+n+3)}$

PART A – QUESTIONS AND ANSWERS

1. **Define Curvature of a curve.**

   Solution: The rate of bending of a curve in any interval is called the curvature of the curve in that interval. The curvature of a straight line is zero.

2. **Write the formula for radius of curvature of a curve.** **(Jan - 2010)**

   Solution: Radius of curvature in Cartesian coordinates $\rho = \left[\frac{\left(1+(y_1)^2\right)^{3/2}}{y_2} \right]$

3. **Write the formula for radius of curvature in polar coordinates.**

   Solution: Radius of curvature in polar coordinates $\rho = \left[\frac{\left(r^2+(r')^2\right)^{3/2}}{r^2 - rr'' + 2(r')^2} \right]$

4. **Find ρ at any point (1, 1) on the curve xy = c^2.** **(Apr - 2010)**

   Solution: Given xy = c^2

   Differentiating with respect to x, we get

$$x\frac{dy}{dx} + y = 0 \qquad \Rightarrow \quad x\frac{dy}{dx} = -y$$

$$\frac{dy}{dx} = -\frac{y}{x} \qquad \Rightarrow \quad \left(\frac{dy}{dx}\right)_{(1,1)} = -1$$

$$\frac{d^2y}{dx^2} = -\left[\frac{x\left(\frac{dy}{dx}\right) - y(1)}{x^2} \right] \quad \Rightarrow \quad = \quad -\left[\frac{1(-1)-1(1)}{1} \right]$$

$$\left[\frac{d^2y}{dx^2}\right]_{(1,1)} = 2$$

We know that $\rho = \left[\dfrac{\left(1+(y_1)^2\right)^{3/2}}{y_2} \right] = \left[\dfrac{\left(1+(-1)^2\right)^{3/2}}{2} \right]$

$$\rho = \left[\frac{(2)^{3/2}}{2} \right] = \frac{2\sqrt{2}}{2} = \sqrt{2}$$

5. Find the radius of curvature at (x,y) for the curve $a^2 y = x^3 - a^3$ (May -2012)

Solution: Given that $a^2 y = x^3 - a^3$

Differentiating with respect to x, we get

$$a^2 y_1 = 3x^2 \quad \Rightarrow \quad y_1 = \frac{3x^2}{a^2}$$

$$y_2 = \frac{3}{a^2}(2x) \quad \Rightarrow \quad y_2 = \frac{6x}{a^2}$$

We know that $\rho = \left[\dfrac{\left(1+(y_1)^2\right)^{3/2}}{y_2} \right]$

$$\rho = \left[\frac{\left(1+\left(\frac{3x^2}{a^2}\right)^2\right)^{3/2}}{\frac{6x}{a^2}} \right] \quad \Rightarrow \quad \left(1+\frac{9x^4}{a^4}\right)^{3/2}\left(\frac{a^2}{6x}\right)$$

$$\rho = \left(\frac{a^4+9x^4}{a^4}\right)^{3/2}\left(\frac{a^2}{6x}\right) = \frac{\left(a^4+9x^4\right)^{3/2}}{6xa^4}$$

6. Find the radius of curvature at x=1 on $y=\frac{\log x}{x}$ **.**

Solution: Given that $y=\frac{\log x}{x}$

Differentiating with respect to x, we get

$$y_1=\frac{1-\log x}{x^2}$$

$$y_2=\frac{x^2\left(\frac{1}{x}\right)-(1-\log x)2x}{x^4}=\frac{2\log x-3}{x^3}$$

$$y_1 \text{at}(x=1)=1$$

$$y_2 \text{at}(x=1)=-3$$

We know that $\rho=\left[\frac{\left(1+(y_1)^2\right)^{3/2}}{y_2}\right]$

$$\rho=\left[\frac{\left(1+(1)^2\right)^{3/2}}{-3}\right]=\frac{2\sqrt{2}}{3}$$

7. Find the radius of curvature at the point $\left(at^2,2at\right)$ **on the parabola** $y^2=4ax$

Solution: Given that $x=at^2$, $y = 2at$

$$x_1=\frac{dx}{dt}=2at \quad y_1=\frac{dy}{dt}=2a$$

$$x_2=\frac{d^2x}{dt^2}=2a \quad y_2=\frac{d^2y}{dt^2}=0$$

$$\rho=\left[\frac{\left(x_1^{\,2}+y_1^{\,2}\right)^{3/2}}{x_1y_2-y_1x_2}\right]=\left[\frac{\left(4a^2t^2+4a^2\right)^{3/2}}{0-4a^2}\right]$$

$$= \left[\frac{\left(4a^2\right)^{3/2}\left(t^2+1\right)^{3/2}}{-4a^2} \right]$$

$$\rho = -2a\left(t^2+1\right)^{3/2}$$

8. Define circle of curvature.

Solution: The circle whose centre is C($\overline{x}$, $\overline{y}$) and radius ρ is called circle of curvature. The equation is $\left(x-\overline{x}\right)^2 + (y-\overline{y})^2 = \rho^2$.

9. Find the curvature of a circle $x^2+y^2+4x+6y-3=0$. (Jan - 2011)

Solution: Given that $x^2+y^2+4x+6y-3=0$

$$(x+2)^2+(y+3)^2-13=3$$

$$(x+2)^2+(y+3)^2=16$$

$$\therefore \quad \rho = 4$$

Curvature is $\dfrac{1}{\rho} = \dfrac{1}{4}$

10. Define Evolute and Involute of a curve. (Jan- 2011)

Solution: The locus of centre of curvature for a curve is called evolute. If a curve B is the evolute of a curve A, then A is said to be an Involute of the curve B.

11. State two properties of Evolute of the curve. (Jan - 2011)

Solution:

i) The normal at any point of the curve touches the evolute at the corresponding centre of curvature.

ii) The length of an arc of the evolute is equal to the difference between the radii of curvature at the points on the original curve corresponding to the extremities of the arc.

iii) There is one evolute but it has infinite number of involutes.

12. Write the formula for centre of curvature.

Solution: The center of curvature is given by $\bar{x} = x - \frac{y_1}{y_2}(1+y_1)^2$,

$$\bar{y} = y + \frac{(1+y_1)^2}{y_2}.$$

13. What is "Curvature of a circle" at any point on it?

Solution: The curvature of a circle at any point on it is the same and is equal to the reciprocal of its radius.

14. Explain the centre of curvature, circle of curvature and radius of curvature.

Solution: Consider a line segment PC, taken from the inward drawn normal to the curve at P. The length of PC is called radius of curvature at P. The point C is called the centre of curvature at P for the curve. The circle whose centre is C and radius ρ is called the circle of curvature at P of the curve.

15. Define Gamma function. **(May- 2011)**

Solution: The definite integral $\int_0^\infty e^{-x}x^{n-1}dx$ exists only when n > 0 which is a function of n and is called Gamma function. It is denoted by $\Gamma(n)$. (i.e.)

$$\Gamma(n) = \int_0^\infty e^{-x}x^{n-1}dx$$

16. Define Beta function. **(May- 2012)**

Solution: The definite integral $\int_0^1 (1-x)^{n-1}x^{m-1}dx$ exists only when n > 0, m>0 which is a function of m and n and is called Beta function. It is denoted by $\beta(m,n)$

(i.e.) $\beta(m,n) = \int_0^1 (1-x)^{n-1}x^{m-1}dx$.

17. Write the relation between Gamma and Beta function. **(Apr- 2010)**

Solution: $\beta(m,n)=\dfrac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$

18. Define Improper integral. Give example for each kind of improper integral.

Solution: The integral $=\int_a^b f(x)dx$ is called an improper integral if any one of the following condition is satisfied.

i) $a=-\infty$ or $b=\infty$ or both. (i.e) one or both the limits of integration is infinite.

ii) f(x) is unbounded at one or some points in the interval $a\le x\le b$. Integrals corresponding to (i) and (ii) are called improper integrals of first and second kind respectively. Integrals with both the conditions (i) and (ii) are called improper integrals of third kind.

Examples

i) $\int_0^\infty \dfrac{dx}{1+x^2}$ is an improper integral of I kind.

ii) $\int_1^0 \dfrac{dx}{1-x^2}$ is an improper integral of II kind.

iii) $\int_0^\infty e^{-x}x^{n-1}dx$ when n > 0 is an improper integral of III kind.

19. Express Beta function in three differential forms.

Solution: 1. $\beta(m,n)=\int_0^1 (1-x)^{n-1}x^{m-1}dx \quad for\ m,n>0$

2. $\beta(m,n)=\int_0^\infty \dfrac{y^{n-1}}{(1+y)^{m+n}}dy$

3. $\beta(m,n)=\int_0^{\pi/2} \sin^{2m-1}\theta\cos^{2n-1}\theta\ d\theta$

20. Is $\beta(m,n)$ Symmetric. Why?

Solution: Yes. $\beta(m,n)$ is a symmetric function as we can P.T. $\beta(m,n)=\beta(n,m)$

We know that $\beta(m,n)=\int_0^1 (1-x)^{n-1}x^{m-1}dx$

By using the property $\int_0^a f(x)dx=\int_0^a f(a-x)dx$

$$\beta(m,n)=\int_0^1 (1-x)^{m-1}x^{n-1}dx \;=\beta(n,m)$$

21. Derive the reduction formula for $\Gamma(n)$.

Solution: By definition, $\Gamma(n+1)=\int_0^\infty e^{-x}x^n\,dx$

Integration by parts we have,

$$\Gamma(n+1)=\left[-x^n e^{-x}\right]_0^\infty+n\int_0^\infty e^{-x}x^{n-1}dx$$

$$=n\int_0^\infty e^{-x}x^{n-1}dx=n\Gamma(n)$$

22. Prove that $\Gamma(n+1)=n!$, n is a positive integer.

Solution: By reduction formula for $\Gamma(n)$, $\Gamma(n+1)=n\Gamma(n)$, when n is a positive integer, $\Gamma(n+1)=n\Gamma(n)$

$$\Gamma(n+1)=n\Gamma(n-1+1)$$
$$\Gamma(n+1)=n(n-1)\Gamma(n-1)$$
$$\Gamma(n+1)=n(n-1)(n-2)\Gamma(n-2)$$

. $\ldots$

$\ldots$

$$\Gamma(n+1)=n(n-1)(n-2)...\Gamma(1)$$
$$\Gamma(n+1)=n(n-1)(n-2)...1$$
$$\Gamma(n+1)=n!$$

23. Prove that $\Gamma(1) = 1$.

Solution: We know that $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx$

Put n = 1

$$\Gamma(1) = \int_0^\infty e^{-x} x^0 dx \int_0^\infty e^{-x} dx = -\left[e^{-x}\right]_0^\infty = -\left[e^{-\infty} - e^0\right] = 1$$

24. Prove that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.

Solution: We know that $\int_0^{\pi/2} \sin^p x \cos^q x dx = \frac{1}{2}\left[\frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{\Gamma\left(\frac{p+q+2}{2}\right)}\right]$

Put p = 0 and q = 0

$$\int_0^{\pi/2} dx = \frac{1}{2}\left[\frac{\Gamma\left(\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma(1)}\right]$$

$$[x]_0^{\frac{\pi}{2}} = \frac{1}{2}\left[\Gamma\left(\frac{1}{2}\right)\right]^2 \qquad (\because \Gamma(1) = 1)$$

$$\frac{\pi}{2} = \frac{1}{2}\left[\Gamma\left(\frac{1}{2}\right)\right]^2 \qquad \left(\because \left[\Gamma\left(\frac{1}{2}\right)\right]^2 = \pi\right)$$

$$\left[\Gamma\left(\frac{1}{2}\right)\right] = \sqrt{\pi}$$

25. Write the duplication formula of Gamma function.

Solution: $\Gamma(2m) = \dfrac{2^{2m-1}\Gamma(m)\Gamma\left(m+\frac{1}{2}\right)}{\sqrt{\pi}}$

26. Express the following integrals in terms of Gamma function

a) $\int_0^{\pi/2} \sin^p x \cos^q x \, dx$ **b)** $\int_0^{\pi/2} \sin^p x \, dx$ **c)** $\int_0^{\pi/2} \cos^p x \, dx$

Solution: a) $$\int_0^{\pi/2} \sin^p x \cos^q x \, dx = \frac{1}{2} \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{\Gamma\left(\frac{p+q+2}{2}\right)}$$

b) $$\int_0^{\pi/2} \sin^p x \, dx = \frac{\sqrt{\pi}}{2} \frac{\Gamma\left(\frac{p+1}{2}\right)}{\Gamma\left(\frac{p+2}{2}\right)}$$

c) $$\int_0^{\pi/2} \cos^p x \, dx = \frac{\sqrt{\pi}}{2} \frac{\Gamma\left(\frac{p+1}{2}\right)}{\Gamma\left(\frac{p+2}{2}\right)}$$

27. Prove that $\beta(m,n) = \beta(m+1,n) + \beta(m,n+1)$

Proof: R. H. S = $\beta(m+1,n) + \beta(m,n+1)$

$$= \int_0^1 x^m (1-x)^{n-1} dx + \int_0^1 x^{m-1}(1-x)^n dx$$

$$= \int_0^1 x^{m-1}(1-x)^{n-1}[x+1-x]dx = \int_0^1 x^{m-1}(1-x)^{n-1} dx$$

$$= \beta(m,n)$$

28. Compute $\frac{\Gamma(8)}{\Gamma(5)\Gamma(3)}$

Solution: $$\frac{\Gamma(8)}{\Gamma(5)\Gamma(3)} = \frac{7!}{4!2!} = \frac{7.6.5.4!}{4!2!} = 105$$

29. Compute $\dfrac{\Gamma\left(\frac{7}{2}\right)}{\Gamma\left(\frac{3}{2}\right)}$

Solution: Given that $\dfrac{\Gamma\left(\frac{7}{2}\right)}{\Gamma\left(\frac{3}{2}\right)} = \dfrac{\Gamma\left(\frac{5}{2}+1\right)}{\Gamma\left(\frac{1}{2}+1\right)}$

$$= \frac{\frac{5}{2}\Gamma\left(\frac{5}{2}\right)}{\Gamma\left(\frac{3}{2}\right)}. = \frac{\frac{5}{2}\Gamma\left(\frac{3}{2}+1\right)}{\Gamma\left(\frac{3}{2}\right)}$$

$$= \frac{\frac{5}{2}\times\frac{3}{2}\Gamma\left(\frac{3}{2}\right)}{\Gamma\left(\frac{3}{2}\right)} = \frac{\Gamma\left(\frac{7}{2}\right)}{\Gamma\left(\frac{3}{2}\right)} = \frac{15}{4}$$

30. Find the value of $\Gamma(-3.5)$

Solution: By recurrence formula

$$\Gamma(n+1) = n\Gamma(n)$$

$$\Gamma(n) = \frac{\Gamma(n+1)}{n}$$

$$\Gamma(-3.5) = \frac{\Gamma(-2.5)}{-3.5} = \frac{\Gamma(-1.5)}{(-3.5)(-2.5)} = \frac{\Gamma(-0.5)}{(-3.5)(-2.5)(-1.5)}$$

$$= \frac{\Gamma(0.5)}{(-3.5)(-2.5)(-1.5)(-0.5)} = \frac{\sqrt{\pi}}{6.5625} \qquad \left[\because \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}\right]$$